“Open-label multi-center Phase 1/2 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Capivasertib and Selinexor in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)”

EXPORAKT

CLINICAL TRIAL ON MEDICINAL PRODUCT FOR HUMAN USE

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Version no. dated: 1.0 06/03/2024

The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

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SUMMARY

Full title	Open-label multi-center Phase 1/2 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Capivasertib and Selinexor in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)
Acronym/reference	EXPORAKT
Coordinating investigator	*Dr Lina BENAJIBA*
Scientific Director (if applicable)	*Pr Raphaël ITZYKSON*
Sponsor	Assistance Publique – Hôpitaux de Paris
Scientific justification	**R/R AML is an unmet medical need.** Acute myeloid leukemia (AML) is a heterogeneous clonal myeloid neoplasm where abnormal proliferation and impaired differentiation of hematopoietic stem and myeloid progenitor cells impedes normal hematopoiesis. Despite the significant progress made in understanding AML oncogenesis over the past decades, this disease remains devastating with about 18,000 new cases every year in Europe and a five-year overall survival lower than 30%(1). The standard of care (SOC) of newly diagnosed AML includes a '7+3' cytarabine-anthracyclin combination followed by repeated courses of high dose cytarabine with or without allogeneic hematopoietic stem cell transplantation (HSCT)(2). Up to one third of unselected AMLs are refractory to intensive chemotherapy. Outcome of such primary refractory AMLs is dismal, with a median survival shorter than one year (3). Half of patients achieving a remission after intensive therapy eventually relapse. The majority of relapsed AMLs lack tractable FLT3 or IDH1-2 driver mutations and their treatment rely on intensive salvage therapy, with a minority of patients proceeding to a potentially curative HSCT (4). Patients with newly diagnosed AML too old or frail for intensive chemotherapy receive less intensive therapies based on hypomethylating agents (HMAs), which, as single agents, are not curative (5).Though the prognosis of such patients may improve in the coming decade with the addition of venetoclax to frontline HMAs (6), patients relapsing or refractory to HMA-based therapies also have survival rates lower than 12 months (7). As such, R/R AML represents an unmet medical need. Novel treatment options independent of the conventional chemotherapy and HMA backbones used in the frontline setting are thus necessary. **Selinexor as a single agent is active in AML.** Selinexor is a selective orally bioavailable inhibitor of the karyopherin XPO1/CRM1 that facilitates nuclear export of over 200 proteins, including tumor suppressors, transcription factors and cell cycle regulators. Selinexor was recently granted FDA accelerated approval for relapsed/refractory multiple myeloma treatment (NCT02336815) (8). In AML, selective inhibitors of nuclear export (SINE) have shown potent anti-leukemic effects in multiple *in vitro* and *in vivo* models (9–11). Selinexor was first evaluated clinically in R/R AML and unfit elderly patients in a dose-escalation phase I study, showing modest efficacy as a single agent (9% CR/CRi, median OS: 2.7 months, NCT01607892) (12). The recommended phase 2 dose was 60 mg flat dose (~35 mg/m^2) twice

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weekly. The activity of Selinexor was also assessed in combination with the HMA decitabine (32% CR/CRi, median OS 5.9 months, NCT02093403)(13), and with the standard of care “3+7” chemotherapeutic regimen (53% CR/CRi, median OS: 10.3 months, NCT02403310)(14). Both studies confirmed the tolerability profile of Selinexor in an AML population. The most common grade ≥ 3 adverse events (AEs) were febrile neutropenia, diarrhea, hyponatremia, thrombocytopenia, anemia and fatigue. Possible class-specific, unusual AEs included hyponatremia and blurred vision, but these were mostly grade 1-2 and self-limiting. Overall, these trials validated the safety and tolerability of Selinexor in AML patients, both as a single agent or in combination with AML SOCs. They prompt further clinical research efforts towards more clinically effective novel therapeutic combination studies.

Capivasertib is safe in patients with solid tumors.
AKT is encoded by three closely related genes (*AKT1*, *AKT2* and *AKT3*) and is the central node of the PI3K-AKT-mTOR network, at the interface of oncogenic signaling and cancer metabolism (15). AKT is frequently activated in solid tumors, thus representing a promising target (15). Capivasertib is a potent, selective orally bioavailable ATP-competitive inhibitor of all 3 AKT isoforms (pan-AKT inhibitor).
Capivasertib was evaluated in a dose escalation phase I trial in patients with advanced solid tumors (NCT01226316)(16). The recommended phase II dose (RP2D) was 480mg twice daily for 4 days followed by 3 days off within each week. The most frequently reported grade ≥ 2 AEs were hyperglycemia, diarrhea and nausea. Most AEs were managed with supportive care or dose reductions and no deaths related to Capivasertib were reported. Capivasertib has since been evaluated in combination with standard of care therapeutic agents in multiple phase I/II studies in solid tumors with frequent PI3K/AKT pathway activation such as breast and prostate cancer (NCT01226316, NCT02338622, NCT02525068)(17–19). These studies confirmed the tolerability profile of Capivasertib. Several phase III studies are ongoing to evaluate capivasertib efficacy in breast and prostate cancer (NCT03997123, NCT04493853). There is currently no clinical data regarding administration of capivasertib in AML patients, or regarding combination of capivasertib with SINEs such as Selinexor, regardless of disease.

AKT inhibitors and selinexor show strong synergism in multiple pre-clinical models of AML.
Using unbiased large scale functional genomic and proteomic screening approaches, inhibition of AKT was identified as a strong sensitizer to Selinexor in AML cell lines (PIs KC Wood, Duke University Durham, NC & A. Puissant, INSERM, Paris France) (20,21). Mechanistically, exposure of AML cells to single-agent selinexor was shown to induce an on-target, pro-oncogenic activation of the PI3K/AKT signaling, thus mitigating the anti-leukemic activity of Selinexor. Such AKT activation driven by Selinexor exposure contrasts with the low basal AKT levels in AML(22), and creates a therapeutic window for a synthetic lethal combination of Selinexor with pan-AKT inhibitors. Indeed, a strong synergism between Selinexor and pan-AKT inhibitors was observed in a panel of 5 AML cell lines with diverse genetic backgrounds, including difficult to treat subsets (eg *KMT2A* fusions, *FLT3* mutations). This was confirmed with ex vivo treatment of cells from a patient-derived xenograft (PDX) model harboring high-risk

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	mutations (*RUNX1*, *ASXL1*, *EZH2*, *TET2*) reminiscent of secondary-type AML, a frequent and challenging subtype of AML of the elderly(23). The synergistic anti-leukemic activity of pan-AKT inhibitor/selinexor combination was validated *in vivo* in multiple AML models, including xenotransplanted cell lines, syngeneic and PDX mouse models. In all models, combined treatment with Selinexor 15 mg/kg and the pan-AKT inhibitor Ipatasertib 65 mg/kg every other day for a total of 4 or 5 doses in mice with overt disease combination significantly reduced leukemic burden and improved survival compared to either drug alone or vehicle. Of note, both drugs showed limited but detectable single-agent activity over vehicle, though the magnitude of Ipatasertib potency was lower than that of Selinexor at the tested concentrations, and well below that of the combination. Importantly, the anti-leukemic effect of this novel combination was superior to the standard of care AML chemotherapeutic regimen in an aggressive syngeneic AML mouse model driven by the *Mll-Af9* fusion. Compared to a '5+3' regimen of cytarabine-anthracyclin, representing the MTD of SOC chemotherapy in this AML model, pan-AKT inhibitor / Selinexor combination led to prolonged survival. Finally, in a secondary transplantation study in limit dilution, pan-AKT inhibition combined to Selinexor led to a ~30-fold reduction in relapse-initiating leukemic stem cells over this SOC chemotherapy regimen. **Collectively, these pre-clinical findings provide a strong rationale for the clinical evaluation of Capivasertib-Selinexor combination in R/R AML patients.** The purpose of this phase 1/2 study is to evaluate the safety and feasibility of such strategy, provide preliminary signals of efficacy, and identify potential biomarkers.
Main objective and primary endpoint	**Main Objectives** Phase 1: To assess the safety of the combination of Capivasertib (Capiva) and Selinexor (Sel) in patients with R/R AML, characterizing the dose-limiting toxicities (DLTs), and identifying the maximal tolerated doses combination (MTDC) and recommended phase 2 doses combination. Phase 2: To assess preliminarily anti-leukemia efficacy of the combination of Capiva and Sel, in R/R AML with reference from historical data on single-agent Selinexor activity (lower threshold for meaningful activity 20% ORR)(12) while monitoring the safety profile of the treatment. **Primary endpoint**: Phase 1: Dose Limiting Toxicity (DLT) DLT will be defined during a safety observation period corresponding to the first cycle of 28 days of Capiva/sel treatment, or within 42 days of the first dose of study treatment (for Grade ≥ 3 neutropenia and thrombocytopenia only), as: ● Prolonged treatment-related Grade ≥ 3 neutropenia and/or Grade ≥ 3 thrombocytopenia lasting for 6 weeks or more from start of therapy without evidence of leukemia or underlying myelodysplastic syndrome ● Grade ≥ 3 non- hematological toxicity with the exception of: o Grade 3 or 4 infection o Grade 3 or 4 fever with neutropenia

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	○ Grade 3 nausea, vomiting or diarrhea that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment ○ Grade 3 hyponatremia that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment ○ Grade 3 hyperglycemia that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment ○ Grade 3 hyperlipidemia that can be managed to ≤ Grade 2 within ≤ 2 weeks of symptomatic treatment ○ Grade 3 tumor lysis syndrome that improves to ≤ Grade 2 within 72 hours of onset with medical treatment ○ Any Grade ≥ 3 asymptomatic laboratory abnormalities that improves to ≤ Grade 2 within 72 hours ○ Anorexia or fatigue RP2DC can be equal to or one dose level lower than a MTDC. The choice of Combination Dose Level will depend on DSMB recommendations based on safety data (1 or >1 combinations recommended by the dose finding design), PK/PD data and efficacy signals from Phase 1. <u>Phase 2: Co-primary endpoints</u> - <u>Overall response rate</u> at 28 days according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations)(26), defined as any of the following: complete response (CR, CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL),CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL)), MLFS (Morphologic leukemia-free state: blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery), and Partial response (PR): absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage to 5 to 25% and decrease of pretreatment bone marrow blast percentage by at least 50%). - <u>Dose Limiting Toxicity</u> (DLT), within 28 days, as defined above
Secondary objectives and endpoints	**Secondary objectives** For the combination of Capivasertib (Capiva) and Selinexor (Sel) in patients with R/R AML, - To characterize the safety profile, in terms of adverse events - To characterize the pharmacokinetics (PK) of both Capiva and Sel when administered in combination, during the dose escalation phase 1. - To assess preliminarily the anti-leukemia efficacy in terms of response - To assess the duration of response - To assess early mortality - To assess the event free survival - To assess the relapse free survival - To assess overall survival

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- To assess subsequent allogeneic HSCT, overall and in responding patients specifically
- To assess molecular Residual Disease (MRD) in patients achieving CR, CRi or CRh

Secondary endpoints

- Adverse events (AE), treatment emergent adverse events (TEAE) and treatment-related TEAE will be evaluated according to the NCI CTCAE version 5.0. Serious adverse events will be defined per protocol according to EMA/ICH criteria.

Count per patient, duration and maximal severity of each TEAES, treatment related TEAE, TEAES leading to treatment discontinuation, serious adverse events (SAE) and deaths

- Pharmacokinetics in phase I patients: Sel plasma levels from blood samples collected at D1C1 (pre-dose, 0.5, 2, 6 and 24 hours post-treatment), D3C1 (pre-dose, 0.5, 2, 6 hours post-treatment), in
- Pharmacokinetics in phase I patients: Capiva plasma levels from blood samples collected at D1C1 (pre-dose, 0.5, 2, 6 and 24 hours post-treatment), D3C1 (pre-dose, 0.5, 2, 6 hours post-treatment).
- Response distribution at day 28 defined according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations) (26):
    - complete remission (CR),
    - CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL)
    - CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL),
    - morphologic leukemia-free state (MLFS), blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery.
    - Partial response (PR, absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage to 5 to 25% and decrease of pretreatment bone marrow blast percentage by at least 50%),
- Duration of response (DOR) up to 24 months, defined as the time from onset of response to progression or death due to any reason, whichever occurs earlier
- All-cause mortality within 28 days (after 1 cycle of treatment) and 56 days (after 2 cycles)
- Event-free survival (EFS) up to 24 months, with events defined as:
    - Non achievement of hematologic response including CR, CRi, CRh
    - Hematologic relapse
    - Initiation of any subsequent anti-leukemic therapy (excluding hydroxyurea)
    - Deaths (any cause) prior to one of the events mentioned above

Reasons for subsequent therapy initiation (including intolerance, resistance, hematologic relapse, unsatisfactory MRD evolution), its type

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	(intensive, less-intensive, best supportive care) and its frame (clinical trial, off-label or compassionate use) will be prospectively recorded. - Relapse free survival (RFS) up to 24 months (defined only for patients achieving CR, CRh, or CRi; measured from the date of achievement of remission until the date of hematologic relapse or death from any cause) - Overall survival (OS) up to 24 months - Cumulative incidence of subsequent allogeneic HSCT up to 24 months, overall and in responding patients specifically - For patients achieving CR, CRi or CRh, molecular Residual Disease (MRD) at day 28 and 56 utilizing Peripheral blood and BM RQ-PCR for NPM1 patients(24), BM RQ-PCR for t(8;21) and inv(16) patients(25), otherwise BM FCM. Apart from specific mentions, all these haematological endpoints will be defined according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations)(26). **Exploratory objectives** To assess the combined pharmacodynamics of Capiva/Sel in leukemia cells of peripheral blood and bone marrow, during both phases of the clinical trial. To identify potential biomarkers of response. **Exploratory endpoints** 1) <u>Pharmacodynamics</u> ● XPO1 mRNA and protein expression by RQ-PCR and Western Blot of PBMCs/BMNCs ● P2RY2 and TP53 mRNA expression by RQ-PCR of PBMCs/BMNCs, ● AKT protein expression and phosphorylation levels of threonine 308 (Thr308) and serine 473 (Ser473) by Western-Blot. pAKT Thr308 levels will also be evaluated by flow-cytometry in BMNCs ● pS6, pPRAS40 by Western-Blot in BMNCs ● FOXO nuclear localization in BMNCs ● phosphorylated GSK3β (pGSK3β) in PRP ● insulin, glucose, and c-peptide levels in blood samples ● exploratory: RPPA in selected patients (responders/non responders) in BMNCs 2) <u>Biomarkers of response</u> • Cytogenetic analysis • Targeted sequencing including recurrent AML drivers and PI3K pathway genes
Design of the study	Phase 1/2, open label, multicenter clinical trial with two subsequent phases: **Phase 1:** dose-escalation with a Bayesian adaptive dose escalation design for drug combination, the product of independent beta probabilities dose escalation (PIPE) phase I design (27), to identify the maximal tolerated dose

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	contour and recommended phase 2 dose (RP2D). The time-to-event extension of the PIPE design may be used depending on the anticipated accrual rate compared to the DLT observation window (observation window 28 days) (28). Phase 1 will be conducted in patients with R/R AML agnostic of genetic mutation status. **Phase 2:** Extension at the MTDC or at the RP2DC chosen by the data and safety monitoring board (DSMB), to confirm the safety of RP2DC and preliminarily document its clinical efficacy. A Bayesian adaptive design will be used, assessing response probability (overall response rate) and monitoring toxicity (29). The design will include futility and safety decision rules (interim and final). Inclusion will be stratified on the *TP53* status to ensure a maximum 20% patient with mutated *TP53* (n=4 for a phase 2 sample of 20 patients).
Category	*Cat 1: Phase 1*
Population of study participants	Patients aged 18 years old or more with R/R AML
Inclusion criteria	● Patients aged 18 years old or more ● With confirmed diagnosis of R/R AML (frontline treatment options include 7+3, GO-based induction, CPX-351, AZA, AZA±Venetoclax, non-exhaustive list) ● Failure or ineligibility for usual salvage therapy (AML) ● WBC < 50,000/mm3 ● ECOG performance status ≤2 ● Any prior treatment-related toxicities resolved to ≤ Grade 2 prior to first study dose, with the exception of alopecia ● AST and ALT ≤3.0 x upper the limit of normal (ULN) and total and direct serum bilirubin ≤ 1.5 x ULN (or ≤2.5 ULN for patients with Gilbert's syndrome), unless considered due to leukemia ● For the inclusion in the phase 2 part: *TP53* mutation status available ● Estimated glomerular filtration rate (GFR) ≥ 50 mL/min according to the MDRD equation ● Written informed consent obtained prior to any screening procedures ● Eligible for National Health Insurance in France
Exclusion criteria	● Acute promyelocytic leukemia ● Proven central nervous system leukemic involvement ● Myeloid Sarcoma with < 20% bone marrow blasts ● Prior allogeneic HSCT within the last 12 weeks or donor lymphocyte conditioning (DLI) within the last 4 weeks ● Graft-Versus-Host Disease (GVHD): Signs or symptoms of acute GVHD ≥ Grade 2, moderate or severe limited chronic GVHD or extensive chronic GVHD of any severity within the last 4 weeks prior to the first study dose. Patients must have been off all systemic immunosuppressive therapy for at least 2 weeks prior to enrollment. Patients may receive as GVHD treatment corticosteroid doses ≤ 10 mg of prednisone equivalent daily. ● Concurrent therapy with any anti-cancer therapy within 14 days or 5 half-lives (whichever is shorter) before the first study dose. Only hydroxyurea for the control of blood counts is permitted. ● Uncontrolled diabetes defined as HbA1c > 9.5% ● Diabetes requiring insulin treatment ● Grade ≥ 2 hypercholesterolemia or hypertriglyceridemia

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	● Adults weighting less than 40 kg ● Uncontrolled systemic fungal, bacterial, or viral infection (defined as ongoing signs/symptoms related to the infection without improvement despite appropriate treatment) ● HBV or HCV hepatitis with a detectable viral load or any ongoing anti-viral treatment ● HIV infection ● Other uncontrolled or active malignant disease within prior 12 months (excluding myelodysplastic syndrome; myeloproliferative neoplasm, cutaneous basal cell carcinoma, “in-situ” carcinoma of the cervix or breast, or other local malignancy excised) ● Persistent (3 consecutive ECGs performed ≥5 minutes apart) prolongation of the corrected QT interval (QTcF) to > 480 msec ● Inability to swallow ● Known malabsorption syndrome or other condition that may significantly impair absorption of oral study medications ● Participation in another therapeutic interventional clinical study within 30 days of enrollment ● Administration of any therapy that is considered to be investigational (i.e., used for non-approved indications(s) or in the context of a research investigation) within 14 days or 5 drug half-lives (whichever is shorter) prior to the first dose of study drug ● Any concurrent severe and/or uncontrolled medical condition, which could compromise participation in the study ● Adults subjects to a legal protection order or unable to give their consent ● Persons deprived of their freedom by judicial or administrative decision, person hospitalized without their consent by virtues of articles L 3212-1 and L3213-1 and who are not subject to the provisions of article L 1121-8 ● Females who are pregnant or breastfeeding ● For males whose sexual partner is a woman of childbearing potential: unwillingness or inability of the man or woman to use a highly effective contraceptive method for the entire treatment period and for at least 6 months after completion of protocol treatment ● For heterosexually active woman of childbearing potential: unwillingness or inability to use a highly effective contraceptive method for the entire treatment period and for at least 3 months after the administration of the final dose of study medication Highly effective contraception methods include: combined (estrogen and progestogen containing) or progestogen only hormonal methods associated with inhibition of ovulation, intra-uterine device, surgical sterilization (including bilateral tubal occlusion, partner’s vasectomy) or sexual abstinence if this is the preferred and usual lifestyle of the patient. Women are not regarded as of childbearing potential if they are post-menopausal (at least 2 years without menses) or are surgically sterile (at least 1 month before enrolment). Female patients must not donate or retrieve, for their own use, ova from the time of screening and throughout the treatment period, and for 3 months after the administration of the final dose of study medication.

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	Female patients must agree not to breastfeed from the time of screening and throughout the protocol period, and for 5 half-lives after the administration of the final dose of study medication. Male patients must not freeze or donate sperm starting at screening and throughout the treatment period and 3 months after the administration of the final dose of study medication.
Investigational medicinal product(s)	Patients will receive Selinexor and Capivasertib *per os* (tablets) in consecutive cycles of 28 days, until disease progression, death or consent withdrawal. The study treatment period will continue for up to 24 cycles after the first dose of study drug of the last enrolled patient. Patients who continue to benefit beyond 24 cycles according to investigator's judgment could be enrolled in a subsequent extended treatment access study as part of a protocol amendment if Astrazeneca and Menarini agree to continue drug delivery for these patients. Selinexor is approved and marketed as treatment of multiple myeloma. Capivasertib is evaluated in phase 3 trials in breast and prostate cancer. Dose Levels for Selinexor and Capivasertib are based on RP2D and available safety data(12,16), in keeping with FDA guidance for the co-development of two drugs(30). Selinexor tablets dosed at 20mg, capivasertib tablets dosed at 80mg, 160mg or 200mg. Selinexor will be given at 40mg, 60mg or 80mg once weekly at D1, 8, 15, and 22 of 28-days cycles. Capivasertib will be given at 240, 320, 400 or 480 mg once daily, D1-4, 8-11, 15-18, 22-25, of 28-days cycles. Eight candidate drug doses combinations will be evaluated: ● Combination DL-1: 40 mg Selinexor, 240mg Capivasertib. ● Combination DL-2: 40 mg Selinexor, 320mg Capivasertib. ● Combination DL-3: 60 mg Selinexor, 320mg Capivasertib. ● Combination DL-4: 60 mg Selinexor, 400mg Capivasertib. ● Combination DL-5: 60 mg Selinexor, 480mg Capivasertib. ● Combination DL-6: 80 mg Selinexor, 320mg Capivasertib. ● Combination DL-7: 80 mg Selinexor, 400mg Capivasertib. ● Combination DL-8: 80 mg Selinexor, 480mg Capivasertib. Dose escalation will be conducted according to a dose finding design with sequential dose assignment with cohorts of size 1 to 3, the number of patients eventually assessed at each dose level depending on the DLT observations.
Comparator treatment	*None*
Interventions added for the study	Interventions added by the study are limited to the following procedures: – BM aspirates including morphology/cytogenetics/genetics at screening, – BM aspirates including morphology and MRD evaluation if CR/CRi/CRh achieved at C2D1, C3D1, every 3 cycles and end of study, – ECG at screening, C1D1, C1D15 and end of study,

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	– Hematology and chemistry laboratory analysis on peripheral blood at screening, C1D1, C1D2, C1D3, C1D8, C1D15, C1D22 and end of study, – Pregnancy test at screening, at each cycle D1 and end of study, – PK (only in the phase 1 part of the study) and PD exams from peripheral blood samples (plasma, serum and cells) and bone marrow samples (cells).
Expected benefits for the participants and for society	Foreseeable benefits are the identification of the most efficient and less toxic dose level combination of Capivasertib/Selinexor. This novel therapeutic strategy should result in improvement of the complete response, overall response rates and long-term outcomes of R/R AML patients.
Practical implementation	Consecutive eligible patients will be included following informed consent. This is a Phase 1/2, open-label, multicenter clinical trial with two consecutive phases (patients will participate in either Phase 1 or Phase 2): Phase 1: Bayesian adaptive dose escalation scheme for the combination of two drugs in patients with relapsed or refractory AML, irrespective of their molecular abnormalities. This part will be multicentric, with 3 CLIP^2-INCa accredited centers. Phase 2: extension at the MTD or recommended phase 2 dose (RP2D) after consulting the independent Data and Safety Monitoring Board (DSMB), to confirm the safety of RP2D and document its clinical anti-leukemic activity on a preliminary basis. This phase will be multicentric (5 centers).
Number of participants included	*Total:44* *Phase 1: 24 patients* *Phase 2: 20 patients*
Number of centres	○ *3 centers in Phase 1* ○ *5 centers in Phase 2* *Conducted only in France, in hematology centers.*
Duration of the study	- *inclusion period: 12 months for the phase 1 and 9 months for the phase 2* - *treatment period: 24 months* - *follow-up period: 24 months* - *total duration: 45 months*
Number of enrolments expected per site and per month	0.2 to 1.1
Statistical analysis	Bayesian adaptive designs will be used for the trial objectives. They allow sequential monitoring of data during the trial, using all current evidence for straightforward inference on safety and efficacy and decision making to select and evaluate an optimal dose. **Phase 1** A product of independent beta probabilities dose escalation phase I design will be used in phase 1, for up to 24 patients (27). Target DLT rate for the

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	definition of the MTD contour (MTC) will be 20%, across the dosing grid including 8 candidate drug doses combinations, defined by 3 candidate dose levels of Selinexor and 4 of Capivasertib (see section Investigational Medicinal Products). A time-to-event extension design may be used depending on the anticipated accrual rate compared to the DLT observation window (observation window 28 days) (28). The design will be calibrated (beta priors, escalation strategy, cohort size and safety constraints) to ensure coherence of the dose escalation scheme and obtain desirable operating characteristics (notably, probability of correct MTC selection, probability of overdose selection, safety of treatment assignments during the trial). **Phase 2** Once phase 1 has been completed providing the estimated MTD, an extension of up to 44 patients will be initiated. Assuming 24 patients have been included in the phase 1 part of the study, 20 new patients will be included in the phase 2 part. The extension will follow a Bayesian adaptive design, assessing the probability of response (overall response rate) and monitoring toxicity (29). The choice of Dose Level for the extension phase will depend on DSMB recommendations based on safety data (1 or >1 combinations recommended by the dose finding design) and efficacy signals from Phase 1. Inclusion criteria will be similar as in Phase 1. Using the Bayesian Optimal Phase 2 design (BOP2) (29), a maximum total of 20 patients, with an interim analysis on the first 9 observations with futility and toxicity decision rules, allows 80% power given a 5% one-sided false positive rate (31), assuming a 10% response rate as unacceptable and 40% as desirable, while a limiting toxicity rate of 40% will be considered as unacceptable and 20 % as acceptable (as in the phase 1 part). We assume independent response and toxicity rates. Among the first 9 patients, if no response has been observed (no. response=0) or if 4 or more limiting toxicities have been observed, the trial will be stopped. Otherwise (if at least 1 response and no more than 3 limiting toxicities), accrual will resume up to a maximum of 20 patients in this part. At the second analysis, we will reject the undesirable hypothesis and conclude to a promising treatment if ≥4 responses are observed overall with ≤6 toxicities. Of note, inclusion in the phase II part will be stratified on the TP53 status to ensure a maximum 20% patient with mutated TP53 (n=4 for a phase 2 sample of 20 patients)
Funding sources	Main study funding: PHRC-I Study medication provided by Menarini (Selinexor) and Astrazeneca (Capivasertib).
Study will have a Data Safety Monitoring Board	*Yes.*

1 SCIENTIFIC JUSTIFICATION FOR THE STUDY

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1.1 Hypothesis for the study

AKT inhibitors and selinexor show strong synergism in multiple pre-clinical models of AML therefore providing a strong rationale for the clinical evaluation of Capivasertib-Selinexor combination in R/R AML patients. This phase 1/2 study hypothesises that the capivasertib-selinexor combination will be safe and feasible, and will result in meaningful overall response rates thus improving AML patients outcome.

1.2 Description of knowledge relating to the condition involved

Acute myeloid leukemia (AML) is a heterogeneous clonal myeloid neoplasm where abnormal proliferation and impaired differentiation of hematopoietic stem and myeloid progenitor cells impedes normal hematopoiesis. Despite the significant progress made in understanding AML oncogenesis over the past decades, this disease remains devastating with about 18,000 new cases every year in Europe and a five-year overall survival of only 17%(1).

The standard of care (SOC) of newly diagnosed AML includes a '7+3' cytarabine-anthracyclin combination followed by repeated courses of high dose cytarabine with or without allogeneic hematopoietic stem cell transplantation (HSCT)(2). Up to one third of unselected AMLs are refractory to intensive chemotherapy. Outcome of such primary refractory AMLs is dismal, with a median survival shorter than one year (3). Half of patients achieving a remission after intensive therapy eventually relapse. The majority of relapsed AMLs lack tractable FLT3 or IDH1-2 driver mutations and their treatment rely on intensive salvage therapy, with a minority of patients proceeding to a potentially curative HSCT (4). Patients with newly diagnosed AML too old or frail for intensive chemotherapy receive less intensive therapies based on hypomethylating agents (HMAs), which, as single agents, are not curative (5).Though the prognosis of such patients may improve in the coming decade with the addition of venetoclax to frontline HMAs(6), patients relapsing or refractory to HMA-based therapies also have survival rates lower than 12 months (7). As such, R/R AML represents an unmet medical need. Novel treatment options independent of the conventional chemotherapy and HMA backbones used in the frontline setting are thus necessary.

1.3 Summary of relevant pre-clinical experiments and clinical trials

Selinexor as a single agent is active in AML.

Selinexor is a selective orally bioavailable inhibitor of the karyopherin XPO1/CRM1 that facilitates nuclear export of over 200 proteins, including tumor suppressors, transcription factors and cell cycle regulators. Selinexor was recently granted FDA accelerated approval for relapsed/refractory multiple myeloma treatment (NCT02336815) (8). In AML, selective inhibitors of nuclear export (SINE) have shown potent anti-leukemic effects in multiple *in vitro* and *in vivo* models (9–11). Selinexor was first evaluated clinically in R/R AML and unfit elderly patients in a dose-escalation phase I study, showing modest efficacy as a single agent (9% CR/CRi, median OS: 2.7 months, NCT01607892) (12). The recommended phase 2 dose was 60 mg flat dose (~35 mg/m^2) twice weekly. The activity of Selinexor was also assessed in combination with the HMA decitabine (32% CR/CRi, median OS 5.9 months, NCT02093403)(13), and with the standard of care "3+7" chemotherapeutic regimen (53% CR/CRi, median OS: 10.3 months, NCT02403310)(14). Both studies confirmed the tolerability profile of Selinexor in an AML population. The most common grade ≥ 3 adverse events (AEs) were febrile neutropenia, diarrhea, hyponatremia, thrombocytopenia, anemia and fatigue. Possible class-specific, unusual AEs included hyponatremia and blurred vision, but these were mostly grade 1-2 and self-limiting. Overall, these trials validated the safety and tolerability of Selinexor in AML patients, both as a single agent or in combination with AML SOCs. They prompt further clinical research efforts towards more clinically effective novel therapeutic combination studies.

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Capivasertib is safe in patients with solid tumors.
AKT is encoded by three closely related genes (*AKT1*, *AKT2* and *AKT3*) and is the central node of the PI3K-AKT-mTOR network, at the interface of oncogenic signaling and cancer metabolism (15). AKT is frequently activated in solid tumors, thus representing a promising target (15). Capivasertib is a potent, selective orally bioavailable ATP-competitive inhibitor of all 3 AKT isoforms (pan-AKT inhibitor).
Capivasertib was evaluated in a dose escalation phase I trial in patients with advanced solid tumors (NCT01226316)(16). The recommended phase II dose (RP2D) was 480mg twice daily for 4 days followed by 3 days off within each week. The most frequently reported grade ≥ 2 AEs were hyperglycemia, diarrhea and nausea. Most AEs were managed with supportive care or dose reductions and no deaths related to Capivasertib were reported. Capivasertib has since been evaluated in combination with standard of care therapeutic agents in multiple phase I/II studies in solid tumors with frequent PI3K/AKT pathway activation such as breast and prostate cancer (NCT01226316, NCT02338622, NCT02525068)(17–19). These studies confirmed the tolerability profile of Capivasertib. Several phase III studies are ongoing to evaluate capivasertib efficacy in breast and prostate cancer (NCT03997123, NCT04493853). There is currently no clinical data regarding administration of capivasertib in AML patients, or regarding combination of capivasertib with SINEs such as Selinexor, regardless of disease.

AKT inhibitors and selinexor show strong synergism in multiple pre-clinical models of AML.
Using unbiased large scale functional genomic and proteomic screening approaches, inhibition of AKT was identified as a strong sensitizer to Selinexor in AML cell lines (PIs KC Wood, Duke University Durham, NC & A. Puissant, INSERM, Paris France) (20,21). Mechanistically, exposure of AML cells to single-agent selinexor was shown to induce an on-target, pro-oncogenic activation of the PI3K/AKT signaling, thus mitigating the anti-leukemic activity of Selinexor. Such AKT activation driven by Selinexor exposure contrasts with the low basal AKT levels in AML(22), and creates a therapeutic window for a synthetic lethal combination of Selinexor with pan-AKT inhibitors. Indeed, a strong synergism between Selinexor and pan-AKT inhibitors was observed in a panel of 5 AML cell lines with diverse genetic backgrounds, including difficult to treat subsets (eg *KMT2A* fusions, *FLT3* mutations). This was confirmed with ex vivo treatment of cells from a patient-derived xenograft (PDX) model harboring high-risk mutations (*RUNX1*, *ASXL1*, *EZH2*, *TET2*) reminiscent of secondary-type AML, a frequent and challenging subtype of AML of the elderly(23). The synergistic anti-leukemic activity of pan-AKT inhibitor/selinexor combination was validated *in vivo* in multiple AML models, including xenotransplanted cell lines, syngeneic and PDX mouse models. In all models, combined treatment with Selinexor 15 mg/kg and the pan-AKT inhibitor Ipatasertib 65 mg/kg every other day for a total of 4 or 5 doses in mice with overt disease combination significantly reduced leukemic burden and improved survival compared to either drug alone or vehicle. Of note, both drugs showed limited but detectable single-agent activity over vehicle, though the magnitude of Ipatasertib potency was lower than that of Selinexor at the tested concentrations, and well below that of the combination. Importantly, the anti-leukemic effect of this novel combination was superior to the standard of care AML chemotherapeutic regimen in an aggressive syngeneic AML mouse model driven by the *Mll-Af9* fusion. Compared to a '5+3' regimen of cytarabine-anthracyclin, representing the MTD of SOC chemotherapy in this AML model, pan-AKT inhibitor / Selinexor combination led to prolonged survival. Finally, in a secondary transplantation study in limit dilution, pan-AKT inhibition combined to Selinexor led to a ~30-fold reduction in relapse-initiating leukemic stem cells over this SOC chemotherapy regimen.

Collectively, these data highlights that AKT inhibitors and selinexor show strong synergism in multiple pre-clinical models of AML.

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1.4 Description of the population to be studied and justification for this choice of participants

- *Criteria related to the disease*
  - Patients with R/R AML who failed or are ineligible to salvage AML therapy are included in the study, as this population has no therapeutic alternative and could benefit from the capivasertib/selinexor combination. Patients will be recruited agnostic from their genetic background as no biomarker of response or resistance to the combined AKT and XPO-1 inhibition has been identified in pre-clinical studies (20).
  - Patients with the favorable cytogenetic lesion t(15;17) defining acute promyelocytic leukemias are excluded from the study as they require specific therapeutic strategies with chemo-free standard of care (32).
  - AML with central nervous system involvement are excluded as they require additional CNS-directed therapy (intrathecal chemotherapy), and because selinexor and capivasertib diffusion properties in CNS are unknown.
  - Myeloid sarcomas without medullary infiltration are excluded because evaluation of response requires different methods compared to AML.
  - Patients with prior allogeneic HSCT within the last 12 weeks or donor lymphocyte conditioning (DLI) within the last 4 weeks, as well as patients having signs or symptoms of acute GVHD ≥ Grade 2, moderate or severe limited chronic GVHD or extensive chronic GVHD of any severity within the last 4 weeks prior to the first study dose, are not eligible. Patients must have been off all systemic immunosuppressive therapy for at least 2 weeks prior to enrollment. Patients may receive as GVHD treatment corticosteroid doses ≤ 10 mg of prednisone daily equivalent. This will facilitate the safety and efficacy assessments of the capivasertib/selinexor combination.

- Criteria related to the use of capivasertib:
  - Patients with uncontrolled diabetes defined as HbA1c > 9.5%, diabetes requiring insulin treatment, grade ≥ 2 hypercholesterolemia or hypertriglyceridemia and adults weighting less than 40 kg are excluded from the study due to AKT inhibitors metabolic effects. Indeed, the AKT pathway is a key regulator of glucose and lipid energy metabolism (33).

- Organ failures and functional insufficiency
  - Other selection criteria are usual and intended to exclude patients ineligible for a combined therapy or more susceptible to develop well documented adverse reactions (abnormalities of liver functional tests, renal failure).

1.5 Identification and description of the investigational medications

The investigational medications are capivasertib and selinexor. Further details are provided in Section 7 below.

1.6 Description and justification of the dosage, route of administration, administration schedule and treatment duration

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Patients will receive Selinexor and Capivasertib *per os* (tablets) in consecutive cycles of 28 days, until disease progression, death or consent withdrawal. The study treatment period will continue for up to 24 cycles after the first dose of study drug of the last enrolled patient. Patients who continue to benefit beyond 24 cycles according to investigator's judgment could be enrolled in a subsequent extended treatment access study as part of a protocol amendment if Astrazeneca and Menarini agree to continue drug delivery for these patients.

Selinexor is approved and marketed as treatment of multiple myeloma. Capivasertib is evaluated in phase 3 trials in breast and prostate cancer.
Dose Levels for Selinexor and Capivasertib are based on RP2D and available safety data(12,16), in keeping with FDA guidance for the co-development of two drugs(30).

Selinexor tablets dosed at 20mg, capivasertib tablets dosed at 80mg, 160mg or 200mg.

Selinexor will be given at 40mg, 60mg or 80mg once weekly at D1, 8, 15, and 22 of 28-days cycles.
Capivasertib will be given at 240, 320, 400 or 480 mg once daily, D1-4, 8-11, 15-18, 22-25, of 28-days cycles.
Eight candidate drug doses combinations will be evaluated:

- Combination DL-1: 40 mg Selinexor, 240mg Capivasertib.
- Combination DL-2: 40 mg Selinexor, 320mg Capivasertib.
- Combination DL-3: 60 mg Selinexor, 320mg Capivasertib.
- Combination DL-4: 60 mg Selinexor, 400mg Capivasertib.
- Combination DL-5: 60 mg Selinexor, 480mg Capivasertib.
- Combination DL-6: 80 mg Selinexor, 320mg Capivasertib.
- Combination DL-7: 80 mg Selinexor, 400mg Capivasertib.
- Combination DL-8: 80 mg Selinexor, 480mg Capivasertib.

Further information about the investigational medications is given in section 7 below.

1.7 Summary of the known and foreseeable benefits and risks for the Clinical Trial participants

Foreseeable benefits

Foreseeable benefits are the identification of the most efficient and less toxic dose level combination of Capivasertib/Selinexor. This novel therapeutic strategy should result in improvement of the complete response, overall response rates and long-term outcomes of R/R AML patients.

R/R AML represents an unmet medical need with a five-year overall survival lower than 30% (1). First-line AML treatment regimens rely on intensive chemotherapy or hypomethylating agents (HMAs) combines with venetoclax. Up to one third of unselected AMLs are refractory to intensive chemotherapy harbouring a dismal outcome, with a median survival shorter than one year (3). Half of patients achieving a remission after intensive therapy eventually relapse. The majority of relapsed AMLs lack tractable FLT3 or IDH1-2 driver mutations and their treatment rely on poorly efficient intensive salvage therapy, with a minority of patients proceeding to a potentially curative HSCT (4). Patients relapsing or refractory to HMA-based therapies also have survival rates lower than 12 months (7). As such, novel treatment options independent of the conventional chemotherapy and HMA backbones used in the frontline setting are thus necessary.

Disease response will be documented according to ELN 2022 recommendations (26). Long term outcome will be documented in all the patients through the standard assessment of the

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duration of response (DOR), relapse free survival (RFS), overall survival (OS). Though the limited sample size and non-comparable design of this early phase clinical trial will not enable to demonstrate an improvement of these efficacy endpoints, the anti-leukemia efficacy of the combination of Capiva and Sel in R/R AML will be interpreted with reference from historical data on single-agent Selinexor activity (12).

Foreseeable risks

Foreseeable risks are related to the potential toxicity of Selinexor and Capivasertib.

Safety profile of **Selinexor** has been documented in the AML population in three clinical studies, as a single agent (12), in combination with the HMA decitabine (13), or in combination with intensive chemotherapy (14). These studies confirmed the safety and tolerability profile of Selinexor in an AML population both as a single agent or in combination with AML SOCs. The most common grade ≥ 3 adverse events (AEs) were febrile neutropenia, diarrhea, hyponatremia, thrombocytopenia, anemia and fatigue. Possible class-specific, unusual AEs included hyponatremia and blurred vision, but these were mostly grade 1-2 and self-limiting.

There is currently no clinical data regarding administration of **capivasertib** in AML patients. Capivasertib was evaluated in advanced solid tumors as a single agent (16) or in combination standard of care therapeutic agents)(17–19). These studies confirmed the tolerability profile of Capivasertib. The most frequently reported grade ≥ 2 AEs were hyperglycemia, diarrhea and nausea. Most AEs were managed with supportive care or dose reductions and no deaths related to Capivasertib were reported.

There is currently no clinical data regarding combination of capivasertib with SINEs such as Selinexor, regardless of disease. A particular attention will be drawn to potential overlapping toxicities between Selinexor and Capivasertib, with specific recommendations for management of gastro-intestinal symptoms.

Clinical examinations and hematology and chemistry laboratory tests performed during the study will enable monitoring of such expected AEs.

The cautious dose-escalation design described in the **section 1.6** and the Bayesian approach, with a continuous reassessment of safety endpoints enables to limit the risks related to the combination of these two drugs.

Additionally, specific management procedures including dose adaptation and treatment interruptions are planned in the study **(Section 6.1.1).**

Finally, the study does not impose any additional medical procedures in besides those which are indicated for the efficacy and safety monitoring of AML salvage therapy.

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2 OBJECTIVES

2.1 Primary objective

Phase 1:
To assess the safety, characterize the dose-limiting toxicities (DLTs), and identify the maximal tolerated dose combinations (MTDC), and recommended phase 2 dose (RP2D) of the combination of Capivasertib (Capiva) and Selinexor (Sel) in patients with R/R AML.
Probability of DLT should not exceed 20% at day 28, at the end of the first cycle of Capiva/Sel treatment.
The recommended phase 2 doses combination (RP2DC) will also be determined, to be used in the phase II part of the trial.

Phase 2:
To assess preliminarily the anti-leukemia efficacy of the combination of Capiva and Sel, in R/R AML with reference from historical data on single-agent Selinexor activity (lower threshold for meaningful activity 20% ORR) (12), while monitoring the safety profile of the treatment.

2.2 Secondary objectives

For the combination of Capivasertib (Capiva) and Selinexor (Sel) in patients with R/R AML,

- To characterize the safety profile, in terms of adverse events
- To characterize the pharmacokinetics (PK) of both Capiva and Sel when administered in combination, during the dose escalation phase 1
- To assess the response distribution at day 28
- To assess the duration of response up to 24months
- To assess early mortality at day 28 and 56
- To assess the event free survival at 24 months
- To assess the relapse free survival at 24 months
- To assess overall survival at 24 months
- To assess subsequent allogeneic HSCT, overall and in responding patients specifically, at 24 months
- To assess molecular Residual Disease (MRD) in patients achieving CR, CRi or CRh, at 28 days

2.3 Exploratory objectives

To assess the combined pharmacodynamics of Capiva/Sel in leukemia cells of peripheral blood and bone marrow, during both phases of the clinical trial.

To identify potential biomarkers of response.

3 STUDY DESIGN

3.1 Study endpoints

3.1.1 Primary endpoint

Phase 1: Dose Limiting Toxicity (DLT)

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DLT will be defined during a safety observation period corresponding to the first cycle of 28 days of Capiva/sel treatment, as:

- Prolonged treatment-related Grade ≥ 3 neutropenia and/or Grade ≥ 3 thrombocytopenia lasting for 6 weeks or more from start of therapy without evidence of leukemia or underlying myelodysplastic syndrom
- Grade ≥ 3 non- hematological toxicity with the exception of:
    - Grade 3 or 4 infection
    - Grade 3 or 4 fever with neutropenia
    - Grade 3 nausea, vomiting or diarrhea that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment
    - Grade 3 hyponatremia that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment
    - Grade 3 hyperglycemia that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment
    - Grade 3 hyperlipidemia that can be managed to ≤ Grade 2 within ≤ 2 weeks of symptomatic treatment
    - Grade 3 tumor lysis syndrome that improves to ≤ Grade 2 within 72 hours of onset with medical treatment
    - Any Grade ≥ 3 asymptomatic laboratory abnormalities that improves to ≤ Grade 2 within 72 hours
    - Anorexia or fatigue

The recommended phase 2 doses combination (RP2DC) will also be defined from the analysis of DLTs and estimation of the maximum tolerated contour (MTC) and dose combinations (MTDC). The RP2D can be equal to or one dose level lower than the MTD. The choice of Combination Dose Level will depend on DSMB recommendations based on safety data (1 or >1 combinations recommended by the dose finding algorithm), PK/PD data and efficacy signals from Phase 1.

Phase 2: Co-primary endpoints

- **Overall Response Rate** at 28 days according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations)(26), defined as any of the following: complete response (CR, CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL),CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL)), MLFS (Morphologic leukemia-free state: blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery), and Partial response (PR): absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage to 5 to 25% and decrease of pretreatment bone marrow blast percentage by at least 50%).
- Absence of response is defined as patients evaluable for response but not meeting the criteria for CR, CRh, CRi, MLFS or PR.
- **DLT** at 28 days as defined above, for the phase 1 part of the study

3.1.2 Secondary endpoints

- Adverse events (AE), treatment emergent adverse events (TEAE) and treatment-related TEAE will be evaluated according to the NCI CTCAE version 5.0. Serious adverse events will be defined per protocol according to EMA/ICH criteria.

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Count per patient, duration and maximal severity of each TEAES, treatment related TEAE, TEAES leading to treatment discontinuation, serious adverse events (SAE) and deaths

- Pharmacokinetics in phase I patients: Sel plasma levels from blood samples collected at D1C1 (pre-dose, 0.5, 2, 6 and 24 hours post-treatment), D3C1 (pre-dose, 0.5, 2, 6 hours post-treatment),
- Pharmacokinetics in phase I patients: Capiva plasma levels from blood samples collected at D1C1 (pre-dose, 0.5, 2, 6 and 24 hours post-treatment), D3C1 (pre-dose, 0.5, 2, 6 hours post-treatment).
- Response at day 28 defined according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations) (26):
    - complete remission (CR),
    - CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL)
    - CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL),
    - Complete response rate including CR, CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL) and CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL).
    - Morphologic leukemia-free state (MLFS): blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery.

- Partial response (PR): absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage to 5 to 25% and decrease of pretreatment bone marrow blast percentage by at least 50%). Duration of response (DOR) up to 24 months, , defined as the time from onset of response to progression or death due to any reason, whichever occurs earlier
- All-cause mortality within 28 days (after 1 cycle of treatment) and 56 days (after 2 cycles)
- Event-free survival (EFS) up to 24 months, with events defined as:
    - Non achievement of hematologic response including CR, CRi, CRh
    - Hematologic relapse
    - Initiation of any subsequent anti-leukemic therapy (excluding hydroxyurea)
    - Deaths (any cause) prior to one of the events mentioned above

  Reasons for subsequent therapy initiation (including intolerance, resistance, hematologic relapse, unsatisfactory MRD evolution), its type (intensive, less-intensive, best supportive care) and its frame (clinical trial, off-label or compassionate use) will be prospectively recorded.
- Relapse free survival (RFS) up to 24 months (defined only for patients achieving CR, CRh, or CRi; measured from the date of achievement of remission until the date of hematologic relapse or death from any cause)
- Overall survival (OS) up to 24 months
- Cumulative incidence of subsequent allogeneic HSCT up to 24 months, overall and in responding patients specifically
- For patients achieving CR, CRi or CRh, molecular Residual Disease (MRD) at 28 and 56 days Peripheral blood and BM RQ-PCR for NPM1 patients(24), BM RQ-PCR for t(8;21) and inv(16) patients(25), otherwise BM FCM.

Apart from specific mentions, all these haematological endpoints will be defined according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations)(26)

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3.1.3 Exploratory endpoints

1) Pharmacodynamics

- XPO1 mRNA and protein expression by RQ-PCR and Western Blot of PBMCs/BMNCs
- P2RY2 and TP53 mRNA expression by RQ-PCR of PBMCs/BMNCs,
- AKT protein expression and phosphorylation levels of threonine 308 (Thr308) and serine 473 (Ser473) by Western-Blot. pAKT Thr308 levels will also be evaluated by flow-cytometry in BMNCs
- pS6, pPRAS40 by Western-Blot in BMNCs
- FOXO nuclear localization in BMNCs
- phosphorylated GSK3β (pGSK3β) in PRP
- insulin, glucose, and c-peptide levels in blood samples
- exploratory: RPPA in selected patients (responders/non responders) in BMNCs

2) Biomarkers of response

- Cytogenetic analysis
- Targeted sequencing including recurrent AML drivers and PI3K pathway genes

3.2 Description of research methodology

3.2.1 Design of the study

Phase 1/2, open label, multicenter clinical trial with two subsequent phases:

Phase 1: dose-escalation with a Bayesian adaptive dose escalation design for drug combination, the product of independent beta probabilities dose escalation (PIPE) phase I design (27), to identify the maximal tolerated dose contour and recommended phase 2 dose (RP2D). The time-to-event extension of the PIPE design may be used depending on the anticipated accrual rate compared to the DLT observation window (observation window 28 days) (28). Phase 1 will be conducted in patients with R/R AML agnostic of genetic mutation status.

Phase 2: extension at the MTD or at the RP2D chosen by the data and safety monitoring board (DSMB), to confirm the safety of RP2D and preliminarily document its clinical efficacy. A Bayesian optimal adaptive phase II design will be used, assessing response probability (overall response rate) and monitoring toxicity (29). The design will include futility and safety decision rules (interim and final). Of note, inclusion in the phase 2 part will be stratified on the *TP53* mutation status to ensure a maximum of 20% of *TP53* mutated patients (n=4 at most, for n=20 patients in the phase II part).

3.2.2 Number of participating sites

This French multicenter study will involve a total of 3 centers in phase 1 and 5 centers in phase 2.

- **Recruitment centres**

Patients will be recruited in hospitals by hematologists expert in AML management.

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- **Non-recruitment centres / Centralized biology centers**

Flow-based MRD will be carried in the hematology laboratory of Hôpital Saint-Louis (Dr Stéphanie Mathis). Molecular biology MRD in patients with *NPM1* mutations or t(8;21) and inv(16) will be carried locally.

Analysis of PK samples will be done by the clinical pharmacology laboratory of Hôpital Saint-Louis (Dr Lauriane Goldwirt).

Samples for PD will be analyzed at INSERM U1342 under the supervision of Dr Lina Benajiba (Dr Lina Benajiba).

3.2.3 Identification of participants

The participants in this Clinical Trial will be identified as follows:
Site number. (3 digits) - Sequential enrolment number for the site (4 digits) - surname initial - first name initial
This reference number is unique and will be used for the entire duration of the study.

4 IMPLEMENTATION OF THE STUDY

Before any examination or intervention related to the study may be carried out, the investigator must obtain the *freely given, informed and written consent of the participant, or of his/her legal representative* where applicable.

Individuals liable to participate in studies benefit from a preliminary medical examination adapted to the study.

The visits will be conducted by a hematologist investigator expert in acute myeloid leukemia management.

Patients will most often be seen during consultation as an outpatient at screening visit and during the following visits, unless a hospitalization is required to manage adverse events related to the disease or the investigational intervention.

The start of the clinical trial is the inclusion of the first patient.

4.1 Screening visit

The screening visit takes place between *1 day* and no later than *28 days* before the C1D1 visit and is intended to collect the patient informed consent and to assess eligibility criteria.

Specifically, the visit will record:
- Signed informed consent form (ICF)
- Medical and disease history and demographics
- Physical examination
- Height and weight
- Vital signs determination
- ECOG performance status evaluation
- Assessment of prior and concomitant medications, including medications received within 14 days prior to screening

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- 12-lead ECG
- Urinary pregnancy test
- Clinical laboratory tests: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, lipid tests (cholesterol, HDL, LDL, triglycerides), fibrinogen, partial thromboplastin time and prothrombin time [PT]/international normalized ratio [INR]
- Creatinine clearance according to MDRD
- Viral Serology: HIV, HCV, HBV
- Hematological specialized assessment:
    - Bone marrow aspiration and peripheral blood for confirmation of R/R diagnosis and baseline characteristics of the AML. A bone marrow biopsy will be performed only if aspiration inadequate. Bone marrow cytogenetic and routine molecular biology can be performed in local laboratories.
    - Bone marrow aspiration and peripheral blood for determination of FCM LAIP (baseline for flow MRD) sent to the central hematology lab of Saint-Louis hospital.
    - Bone marrow aspiration and peripheral blood for determination of *NPM1* and t(8;21) and inv(16) transcripts quantification (baseline for MRD for *NPM1* mutated or t(8;21) and inv(16) patients) can be performed in local laboratories.
- Biomarkers:
    - Targeted sequencing of a panel of genes recurrently mutated in AML.
    - Targeted sequencing of a panel of genes including PI3K pathway genes.
- PD:
    - Bone marrow aspiration and peripheral blood for: XPO1 mRNA and protein expression by RQ-PCR and Western Blot of PBMCs/BMNCs, P2RY2 and TP53 mRNA expression by RQ-PCR of PBMCs/BMNCs, AKT protein expression and phosphorylation levels of threonine 308 (Thr308) and serine 473 (Ser473) by Western-Blot. pAKT Thr308 levels will also be evaluated by flow-cytometry in BMNCs, pS6, pPRAS40 by Western-Blot in BMNCs, FOXO nuclear localization in BMNCs, phosphorylated GSK3β (pGSK3β) in PRP, insulin, glucose, and c-peptide levels in blood samples, RPPA in selected patients (responders/non responders) in BMNCs

Whose consent must be obtained	**Who informs the individuals and collects their consent**	**At what point the individuals are informed**	**At what point the consent is obtained**
• *the individual participating in the study;* • *the appointed legal representative;*	• *the principal investigator or collaborating physician declared and trained in the study (hematology)*	• *standard of care visit to allow for a minimum of 24h reflection time; information is given again to the patient at the screening visit prior to consent collection*	• *after a reflection period of at least 24h at the time a specific procedure for the study is implemented*

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4.2 Follow-up visits

Visits will take place at C1D1, C1D2, C1D3, C1D8, C1D15, C1D22, C2D1, C2D8, C2D15, C3D1 then every CxD1.

Specifically, the visits will report:

- Physical examination
- Weight
- Vital signs determination
- ECOG performance status evaluation
- Assessment of prior and concomitant medications
- 12-lead ECG (at C1D1 and C1D15)
- Urinary pregnancy test at every CxD1
- Clinical laboratory results: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, lipid tests (cholesterol, HDL, LDL, triglycerides), fibrinogen, partial thromboplastin time and prothrombin time (PT)/international normalized ratio (INR).
- PK (<u>only during phase 1</u>)
    - at C1D1, at pre-dose, 0.5h, 2h, 6h and 24h post-treatment
    - at C1D3 at pre-dose, 0.5h, 2h and 6h post-treatment
- AE/SAE evaluation
- PD at C1D1, C1D2, C1D3 and C2D1:
    - Bone marrow aspiration and peripheral blood for: XPO1 mRNA and protein expression by RQ-PCR and Western Blot of PBMCs/BMNCs, P2RY2 and TP53 mRNA expression by RQ-PCR of PBMCs/BMNCs, AKT protein expression and phosphorylation levels of threonine 308 (Thr308) and serine 473 (Ser473) by Western-Blot. pAKT Thr308 levels will also be evaluated by flow-cytometry in BMNCs, pS6, pPRAS40 by Western-Blot in BMNCs, FOXO nuclear localization in BMNCs, phosphorylated GSK3β (pGSK3β) in PRP, insulin, glucose, and c-peptide levels in blood samples, RPPA in selected patients (responders/non responders) in BMNCs
- Specialized hematological assessment (at C2D1, C3D1 and every 3 cycles):
    - Bone marrow aspiration for assessment of response according to ELN 2022 criteria. A bone marrow biopsy will be performed only if aspirate is inadequate.
    - If CR, CRi or CRh achieved, bone marrow aspiration for flow based MRD (sent to the central hematology lab of Saint-Louis hospital).
    - Bone marrow aspiration and peripheral blood for *NPM1* transcript MRD in *NPM1* mutated patients performed locally.

4.3 Last study visit

- Physical examination
- Weight
- Vital signs determinations
- ECOG performance status evaluation
- Assessment of prior and concomitant medications,
- 12-lead ECG
- Urinary pregnancy test

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- Clinical laboratory results: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, lipid tests (cholesterol, HDL, LDL, triglycerides), fibrinogen, partial thromboplastin time and prothrombin time (PT)/international normalized ratio (INR),
- AE/SAE evaluation
- PD:
    - Bone marrow aspiration and peripheral blood for: XPO1 mRNA and protein expression by RQ-PCR and Western Blot of PBMCs/BMNCs, P2RY2 and TP53 mRNA expression by RQ-PCR of PBMCs/BMNCs, AKT protein expression and phosphorylation levels of threonine 308 (Thr308) and serine 473 (Ser473) by Western-Blot. pAKT Thr308 levels will also be evaluated by flow-cytometry in BMNCs, pS6, pPRAS40 by Western-Blot in BMNCs, FOXO nuclear localization in BMNCs, phosphorylated GSK3β (pGSK3β) in PRP, insulin, glucose, and c-peptide levels in blood samples, RPPA in selected patients (responders/non responders) in BMNCs
- Specialized hematological assessment:
    - Bone marrow aspiration for assessment of response according to ELN 2022 criteria. A bone marrow biopsy will be performed only if aspirate is inadequate.
    - If CR, CRi or CRh achieved, bone marrow aspiration for flow based MRD (sent to the central hematology lab of Saint-Louis hospital).
    - Bone marrow aspiration and peripheral blood for *NPM1* transcript MRD in *NPM1* mutated patients performed locally.

4.4 Post study follow-up

Within 5 years of study completion, investigators may be periodically asked to the following information:

- The vital status of the patient, and if he deceased, the date of his death
- The status of the AML, and, if any, the date of its relapse
- The receipt of allogeneic stem cell transplantation and if any, the date of transplantation.

The patient will be informed and should not oppose to this, as stated in the information note/consent form.

Patients enrolled in an ALFA center will be invited to join the ALFA PPP prospective registry (NCT04777916) and its long-term follow-up cohort. They will have received a separate oral and written information and signed a separate consent form.

4.5 Early termination visit

Patients are allowed to withdraw their participation in the research at any time and for any reason. The investigator can temporarily or permanently withdraw a participant from the study for any safety reason or if it is in the participant's best interests.

If a participant wishes to withdraw their consent from the study, we will use the following strategies to minimize the impact on the validity of the trial meanwhile respecting the

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participant's right to withdraw. We will seek a better understanding of the patient's wishes and offer the following alternatives to complete withdrawal:

1) Discontinue in-person follow-up but allow telephone follow-up;
2) Discontinue in person and by telephone follow-up but allow access to medical records and to use personal data to be able to get life status from the national centre of epidemiology of deaths (Centre d'épidémiologie sur les causes de décès - CépiDc).

In any case, all the data that would have been collected before the patient's consent withdrawal will be kept in. The early termination visit will be similar to the last study visit (see **section 4.4**).

4.6 Expected length of participation and description of the chronology and duration of the study.

Duration of enrolment period	21 months
The length of participation for participants, of which:	
• Maximum period between screening and enrolment:	28 days
• Treatment duration:	24 months
• Duration of follow-up period:	24 months
Total study duration*:	45 months

*End of trial is defined as the last visit of the last patient.

4.7 Table or diagram summarising the chronology of the study

Study Visit	Baseline Screening	Cycle 1						Cycle 2			Cycle 3	Cycle 4+	/ 3 cycles*	Study End
		D1	D2	D3	D8	D15 +/- 2d	D22 +/- 2d	D1 +/- 2d	D8 +/- 2d	D15 +/- 2d	D1 +/- 2d	D1 +/- 2d	D1 +/- 2d	+/- 7d
Informed Consent	x													
Baseline medical history	x													
Physical exam	x	x	x	x	x	x	x	x	x	x	x	x		x
Height, Weight	x	x						x			x	x		x
ECOG PS	x	x						x			x	x		x
BM aspirate[1]	x							x			x		x	x
Cytogenetics[1]	x							x					x	x
Genetics[1]	x							x					x	x
PD[2]	x	x	x	x				x						x
Response Assessment								x			x		x	x
Central MRD[3]	x							(x)			(x)		(x)	(x)
Hematology[4]	x	x	x	x	x	x	x	x	x	x	x	x		x
Chemistry[5]	x	x	x	x	x	x	x	x	x	x	x	x		x
Pregnancy test[6]	x							x			x	x		x
ECG	x	x				x								x
PK[7]		(x)	(x)	(x)										
AE	x	x	x	x	x	x	x	x	x	x	x	x		x

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* in addition of D1 visit in the column left to this one.
[1] Additional BM aspirates with cytomorphology for response assessment (including MRD if CR/CRi/CRh achieved) are also mandatory:
- Before an allogeneic HSCT if any.
- At any time there is a suspicion of progression according to investigator.
[2] PD on peripheral blood, platelet-rich plasma, serum, and BM when visits include BM aspirates.
[3] Central MRD only if CR/CRi/CRh achieved, on peripheral blood and BM RQ-PCR for NPM1 patients(24), BM RQ-PCR for t(8;21) and inv(16) patients(25), otherwise BM FCM.
[4] Including full CBC, fibrinogen, aPTT and PT/INR.
[5] Including glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, BUN, creatinine, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, lipid tests (cholesterol, HDL, LDL, triglycerides).
[6] restricted to women of childbearing potential.
[7] only in the phase 1 part of the study.

4.8 Distinction between standard care and study

TABLE: "Standard care" vs. "additional interventions" required specifically for the study

Interventions, procedures and treatments carried out for research purposes	**Interventions, procedures and treatments associated with <u>standard care</u>**	**Interventions, procedures and treatments added for <u>research purposes</u>**
Treatments		Administration of Selinexor and Capivasertib
Visits	Every 3 months visits	Screening, C1D1, C1D2, C1D3, C1D8, C1D15, C1D22, C2D1, C2D8, C2D15, C3D1, CxD1 and Study exit visit
Blood samples	Monitoring of hematology and chemistry tests every month.	Monitoring of hematology and chemistry tests at screening, C1D2, C1D3, C1D8, C1D15, C1D22, C2D8, C2D15 and Study exit visit. PK monitoring at C1D1 (20 mL), C1D2 (5 mL) and C1D3 (20 mL) for patients in the phase 1 portion of the study. PD monitoring at C1D1 (25 mL), C1D2 (25 mL), C1D3 (25 mL), C2D1 (25 mL) and study exit visit (25 mL).
Urinary test		Pregnancy test at screening, at each cycle D1 and end of study

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Bone marrow aspiration	Every 3 months (6 mL)	Screening (10 mL), C2D1 (10 mL), C3D1 (6 mL), Study end visit (10 mL)
Procedures		ECG at C1D1, C1D15 and Study exit visit

4.9 Biological samples collection

Serum, plasma, nucleic acids, proteins and cells from bone marrow and peripheral blood samples taken at protocol specified visits will be stored in a biological sample collection. The sample will be pseudonymised.
During the study the sample collection(s) will be stored at INSERM U944 under the supervision of Dr Lina Benajiba for no more than 25 years. The collection is intended to perform the biomarker and PD analyses of the study.

At the end of the study, remaining samples will be kept and may be used for further analyses not described in the initial protocol, but which may be useful for investigation of acute myeloid leukemia in light of advances in scientific knowledge, provided the participant is informed and does not oppose this, as stated in the informed consent form.

If the samples are kept at the end of the study, the sample collection will be declared to the ministry of research and to the director of the competent regional healthcare authority - (Article L. 1243-3 of the *Code de la Santé Publique* [French Public Health Code]).

5 ELIGIBILITY CRITERIA

5.1 Inclusion criteria

- Patients aged 18 years old or more
- With confirmed diagnosis of R/R AML (frontline treatment options include 7+3, GO-based induction, CPX-351, AZA, AZA±Venetoclax, non-exhaustive list)
- Failure or ineligibility for usual salvage therapy (AML)
- WBC < 50,000/mm3
- ECOG performance status ≤2
- Any prior treatment-related toxicities resolved to ≤ Grade 2 prior to first study dose, with the exception of alopecia
- AST and ALT ≤3.0 x upper the limit of normal (ULN) and total and direct serum bilirubin ≤ 1.5 x ULN (or ≤2.5 ULN for patients with Gilbert's syndrome), unless considered due to leukemia
- For the phase II part: TP53 mutation status available
- Estimated glomerular filtration rate (GFR) ≥ 50 mL/min according to the MDRD equation
- Written informed consent obtained prior to any screening procedures
- Eligible for National Health Insurance in France

5.2 Exclusion criteria

- Acute promyelocytic leukemia
- Proven central nervous system leukemic involvement
- Myeloid Sarcoma with < 20% bone marrow blasts
- Prior allogeneic HSCT within the last 12 weeks or donor lymphocyte conditioning (DLI) within the last 4 weeks
- Graft-Versus-Host Disease (GVHD): Signs or symptoms of acute GVHD ≥ Grade 2, moderate or severe limited chronic GVHD or extensive chronic GVHD of any severity within the last 4 weeks prior to the first study dose. Patients must have been off all systemic

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immunosuppressive therapy for at least 2 weeks prior to enrollment. Patients may receive as GVHD treatment corticosteroid doses ≤ 10 mg of prednisone equivalent daily.

- Concurrent therapy with any anti-cancer therapy within 14 days or 5 half-lives (whichever is shorter) before the first study dose. Only hydroxyurea for the control of blood counts is permitted.
- Uncontrolled diabetes defined as HbA1c > 9.5%
- Diabetes requiring insulin treatment
- Grade ≥ 2 hypercholesterolemia or hypertriglyceridemia
- Adults weighting less than 40 kg
- Uncontrolled systemic fungal, bacterial, or viral infection (defined as ongoing signs/symptoms related to the infection without improvement despite appropriate treatment)
- HBV or HCV hepatitis with a detectable viral load or any ongoing anti-viral treatment
- HIV infection
- Other uncontrolled or active malignant disease within prior 12 months (excluding myelodysplastic syndrome; myeloproliferative neoplasm, cutaneous basal cell carcinoma, “in-situ” carcinoma of the cervix or breast, or other local malignancy excised)
- Persistent (3 consecutive ECGs performed ≥5 minutes apart) prolongation of the corrected QT interval (QTcF) to > 480 msec
- Inability to swallow
- Known malabsorption syndrome or other condition that may significantly impair absorption of oral study medications
- Participation in another therapeutic interventional clinical study within 30 days of enrollment
- Administration of any therapy that is considered to be investigational (i.e., used for non-approved indications(s) or in the context of a research investigation) within 14 days or 5 drug half-lives (whichever is shorter) prior to the first dose of study drug
- Any concurrent severe and/or uncontrolled medical condition, which could compromise participation in the study
- Adult subjects to a legal protection order or unable to give their consent
- Persons deprived of their freedom by judicial or administrative decision, person hospitalized without their consent by virtues of articles L 3212-1 and L3213-1 and who are not subject to the provisions of article L 1121-8
- Females who are pregnant or breastfeeding
- For males whose sexual partner is a woman of childbearing potential: unwillingness or inability of the man or woman to use a highly effective contraceptive method for the entire treatment period and for at least 6 months after completion of protocol treatment
- For heterosexually active woman of childbearing potential: unwillingness or inability to use a highly effective contraceptive method for the entire treatment period and for at least 3 months after the administration of the final dose of study medication

Highly effective contraception methods include: combined (estrogen and progestogen containing) or progestogen only hormonal methods associated with inhibition of ovulation, intra-uterine device, surgical sterilization (including bilateral tubal occlusion, partner’s vasectomy) or sexual abstinence if this is the preferred and usual lifestyle of the patient.

Women are not regarded as of childbearing potential if they are post-menopausal (at least 2 years without menses) or are surgically sterile (at least 1 month before enrollment).

Female patients must not donate or retrieve, for their own use, ova from the time of screening and throughout the treatment period, and for 3 months after the administration of the final dose of study medication.

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Female patients must agree not to breastfeed from the time of screening and throughout the protocol period, and for 5 half-lives after the administration of the final dose of study medication.

Male patients must not freeze or donate sperm starting at screening and throughout the treatment period and 3 months after the administration of the final dose of study medication.

5.3 Recruitment procedure

Justification of sufficient recruitment capacity for the number of participants that need to be included:
Patients will be recruited by haematologists specialized in AML patients care across 5 centers spread across the country. All 5 centers routinely manage AML patients and given the high rate of refractory / relapsed AML disease (as described in **section 1.2**), we expect to be able to recruit the 44 patients over a period of 19 months (12 months for the phase 1 and 9 months for the phase 2).

	Number of participants
Total number of participants to be included	*A maximum of 44* *24 in phase 1* *20 in phase 2*
Number of centres	*3 in phase 1* *5 in phase 2*
Enrolment period (months)	*12 months for phase 1* *9 months for phase 2*
Number of participants/centre	*3 to 20*
Number of participants/centre/month	***0.2 to 1.1***

5.4 Termination rules

5.4.1 Criteria and procedures for prematurely terminating the study treatment

Several situations are possible
- Temporary suspension of treatment: the investigator must document the reason for suspending and resuming the treatment in the participant's source file and the case report form (CRF)
- Premature discontinuation of treatment, but the participant remains enrolled in the study until the end of their participation
- Premature discontinuation of treatment and withdrawal from the study

The investigator must:
- Document the reason(s)
- Collect the assessment criteria at the time of ending participation in the study, if the participant agrees
- Schedule a follow-up for the participant, particularly in case of a serious adverse event.

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In the case of severe adverse events, the investigator must notify the sponsor and follow up the participant for 1 month following the premature discontinuation of treatment or until the adverse event resolution if it occurred before. Notification of a serious adverse event must be sent by email (eig-vigilance.drc@aphp.fr) to the sponsor. The serious adverse event will be monitored until it is resolved. If a Data Safety Monitoring Board has been created, the committee can specify and/or validate the follow-up methods.

5.4.2 Criteria and procedure for premature withdrawal of a participant from the study

- Participants may withdraw consent, exit the study at any time and for any reason.
- Participant lost to follow-up: the participant cannot be located. The investigator must make every effort to reconnect with the participant (and document his attempts in the source file), at least to determine whether the participant is alive or dead.

If a participant exits the study prematurely or withdraws consent, any data collected prior to the date of premature exit may still be used. Patients who discontinue permanently the Capiva/Sel combination must continue to be followed for documentation of a subsequent leukemia treatment, collection of outcome and/or survival follow-up data until death or conclusion of the study (see above section 5.4.1), unless they withdraw consent for any research procedure and/or use of their data.

- In case of serious adverse events, see the corresponding section on vigilance.

The case report form must list the various reasons why the participant has discontinued the study:

- ☐ Explicit withdrawal of consent
- ☐ Lost to follow-up

5.4.3 Follow-up of participants following premature withdrawal from the study

If a participant discontinues the study, this will in no way affect their usual care for their condition.
In the event of serious adverse events following premature discontinuation of treatment and participation of the patient in the study; see section **5.4.1**

5.4.4 Procedures for replacing participants

Participants in the phase I part of the study who have withdrawn their consent before the end of the evaluation window for reasons other than death or DLT will not be replaced. The dose finding design used allows accounting for incomplete observations due to patients' discontinuation.

Participants in the phase II part of the study who have prematurely discontinued the study before the EOI visit for reasons other than death, lack of efficacy or adverse reaction will be not be replaced. Intent-to-treat set will be used for all main analyses.

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5.4.5 Full or partial discontinuation of the study

Early and temporary or definitive discontinuation of the study, early and temporary discontinuation of inclusions can be decided based on DSMB recommendations, and whether or not participants are allowed to complete their treatment or should immediately discontinue study drugs will all be based on DSMB recommendations. The DSMB will examine all safety data throughout the trial.

AP-HP as sponsor or the Competent Authority can prematurely discontinue all or part of the study, temporarily or permanently, further to the recommendations of the Data Safety Monitoring Board in the following situations:

- first of all, if suspected unexpected serious adverse reactions (SUSARs) are observed, requiring a reassessment of the benefit-risk ratio for the study.
- if an interim analysis confirms the efficacy of the treatment or, alternatively, the lack of efficacy.

Similarly, AP-HP, as the sponsor, or the Competent Authority may decide to prematurely discontinue the study due to unforeseen issues or new information about the product, in light of which the objectives of the study or clinical programme are unlikely to be achieved.

AP-HP as sponsor reserves the right to permanently suspend enrolment at any time if it appears that the inclusion objectives are not met.

If the study is prematurely discontinued for safety reasons, the decision and justification will be provided by the sponsor (AP-HP) to the Competent Authority and to the Ethics Committee without undue delay but not later than in 15 days of the date of the temporary halt or early termination. It shall include the reasons for such action and specify follow-up measures, along with recommendations from the Data Safety Monitoring Board in the case of substantial modification.

6 TREATMENT ADMINISTERED TO STUDY PARTICIPANTS

6.1 Description of the investigational medicinal products

6.1.1 Investigational medicinal products: Selinexor and capivasertib

Patients will receive Selinexor and Capivasertib *per os* (tablets) in consecutive cycles of 28 days, until disease progression, death or consent withdrawal. The study treatment period will continue for up to 24 cycles after the first dose of study drug of the last enrolled patient. Patients who continue to benefit beyond 24 cycles according to investigator's judgment could be enrolled in a subsequent extended treatment access study as part of a protocol amendment if Astrazeneca and Menarini agree to continue drug delivery for these patients.

Selinexor is approved and marketed as treatment of multiple myeloma. Capivasertib is evaluated in phase 3 trials in breast and prostate cancer.
Dose Levels for Selinexor and Capivasertib are based on RP2D and available safety data(12,16), in keeping with FDA guidance for the co-development of two drugs(30).

Selinexor tablets dosed at 20mg, capivasertib tablets dosed at 80mg, 160mg or 200mg.

Selinexor will be given at 40mg, 60mg or 80mg once weekly at D1, 8, 15, and 22 of 28-days cycles.

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Capivasertib will be given at 240, 320, 400 or 480 mg once daily, D1-4, 8-11, 15-18, 22-25, of 28-days cycles.
Eight candidate drug doses combinations will be evaluated:

- Combination DL-1: 40 mg Selinexor, 240mg Capivasertib.
- Combination DL-2: 40 mg Selinexor, 320mg Capivasertib.
- Combination DL-3: 60 mg Selinexor, 320mg Capivasertib.
- Combination DL-4: 60 mg Selinexor, 400mg Capivasertib.
- Combination DL-5: 60 mg Selinexor, 480mg Capivasertib.
- Combination DL-6: 80 mg Selinexor, 320mg Capivasertib.
- Combination DL-7: 80 mg Selinexor, 400mg Capivasertib.
- Combination DL-8: 80 mg Selinexor, 480mg Capivasertib.

Dose escalation will be conducted according to a dose finding design with sequential dose assignment with cohorts of size 1 to 3, the number of patients eventually assessed at each dose level depending on the DLT observations.

Dosing should be interrupted or doses should be reduced as follows:

Adverse events & reactions	Recommended action
Hematological AE grade 4	Continue cycle. Delay D1 of next cycle up to D42 in cycles 1-2. In responding patients beyond cycle 2 (cycle 3+), next cycle can be delayed up to D56 and started at the next lower DL for both Capivasertib and Selinexor.
Non hematological grade ≥3 TEAE	Interrupt Capivasertib and Selinexor until toxicity resolves or improves to grade ≤ 2. Resume Capivasertib and Selinexor at the next lower dose level of both drugs, except for the following TEAEs where treatment can be restarted at the same DL: hyperglycemia, hyperlipidemia, hyponatremia, diarrhea, nausea, vomiting, infection, tumor lysis syndrome. For the latest TEAEs, treatment can be restarted at the next lower dose level if TEAE re-occurs after treatment restart.

For the phase 1 part, no dose adaptation is allowed during the DLT evaluation period (28 days).

Management of GI symptoms
Gastro-intestinal toxicities are expected TEAEs of both Sel and Capiva. They can be managed through symptomatic treatment as follows.

- Non-febrile diarrhea can be managed with loperamide administration starting at 4mg, to be adapted to patient symptoms (maximal daily dose: 16mg). Alternatively, racecadotril can also be used for mild symptoms.
- All patients will be required to use dual anti-emetics prior to each dose of selinexor treatment as nausea prophylaxis for the first two 28-day cycles, with the option to continue dual, single, or no anti-emetics thereafter per investigators decision. The regimen should include a 5-hydroxytryptamine (5-HT3) receptor antagonist, plus one medication with a different mechanism of action. Preferred anti-emetics include the combination of ondansetron 8 mg PO starting with 1 dose the evening before selinexor dose and 30 minutes prior to selinexor and continuing every 8 hours for 2 days after dose plus aprepitant 125 mg 1 hour prior to selinexor, then 80 mg PO daily for 2 days after dose. Patients that are intolerant of 5-HT3 antagonists may combine dual anti-emetics with different mechanisms of action.

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In both cases, patients will be carefully monitored and hydrated orally and/or IV depending on symptoms severity.

Due to the severe, overlapping gastrointestinal (GI) toxicity profiles of both Selinexor and Capivasertib, compliance with the mandated dual anti-emetic prophylaxis (5-HT3 receptor antagonist plus one medication with a different mechanism of action) must be reviewed during the first two cycles.

6.2 Description of Auxiliary medicinal product(s) (treatments required to conduct the study)

According to the Regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014, an auxiliary medicinal product is a medicinal product used in the context of a clinical trial but not as investigational medicinal product.

Not Applicable.

6.3 Authorised and prohibited treatments (medicinal, additional medicinal, surgical), including rescue medications

- Use of other anti-leukemic agents including hypomethylating agents, chemotherapy agents, IDH inhbitors, midostaurine, or other kinase inhibitors, is prohibited.
- Use of erythropoiesis stimulating agents and thrombopoietin analogs is prohibited.
- Administration of strong CYP3A4 inhibitors and inducers: Both Selinexor and Capivasertib are eliminated through CYP3A4 metabolism. If treatment with a strong CYP3A4 inhibitor is required with no available alternative per investigators decision, then a dose reduction of Capiva and Sel to the next lower DL is appropriate. Patients with such concomitant medication will be monitored more closely during the concomitant treatment period. A list of prohibited CYP3A inhibitors or inducers (unless absolutely necessary for patient care) is provided in **ADDENDA 17.6.**
- Allogeneic HSCT might be offered to CR/CRi/CRh patients:
  - Aged ≤ 75 years old
  - Neither transplanted or transplanted more than 1 year before
  - With a composite HSCT co-morbidity index (HSCT-CI)/age less than 5

Stem cell sources will include: HLA-identical siblings; HLA 10/10 matched unrelated donors; HLA 9/10 matched unrelated donors; haplo-identical donors; or cord blood cells (CBC) units (single or double).

Local investigators will choose the conditioning regimen for individual patients. A minimum interval of 7 days should be managed between the last dosing of both Capiva and Sel and HSCT conditioning initiation.

6.4 Methods for monitoring compliance with the treatment

Selinexor and Capivasertib administration will be followed through patients' compliance interrogation by the investigator and tablets counting by the hospital pharmacy at each Cycle Day 1 visit.

7 EFFICACY ASSESSMENT

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7.1 Description of efficacy endpoints assessment parameters

The efficacy endpoints are defined according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations)(26).

- Overall response rate (ORR) being defined as CR, CRi and CRh, MLFS and PR rates:

  - CR is defined as bone marrow blasts < 5% and blasts with Auer rods, absence of circulating blasts; absence of extramedullary disease, ANC ≥ 1.0 x10^9/L and platelets ≥ 100 x10^9/L

  - CRi (CR with incomplete hematologic recovery, is defined as CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL

  - CRh (CR with partial haematological recovery) is defined as CR not fulfilling CR or CRi peripheral blood criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL

  - Morphologic leukemia-free state (MLFS): blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery

  - Partial response (PR): absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage to 5 to 25% and decrease of pretreatment bone marrow blast percentage by at least 50%

- For patients achieving CR, CRi or CRh, molecular Residual Disease (MRD) will be evaluated as follows: Peripheral blood and BM RQ-PCR for NPM1 patients, BM RQ-PCR for t(8;21) and inv(16) patients, otherwise BM FCM.

- Refractory disease defined as patients lacking criteria for any of the above defined responses (CR, CRi, CRh, MLFS and PR).

- Relapsed disease defined for patients achieving CR, CRi, CRh or MLFS as Bone marrow blasts ≥ 5%; or reappearance of blasts in the blood in at least 2 peripheral blood samples at least one week apart; or development of extramedullary disease.

- Outcomes measures

  - Overall survival

  - Event free survival measured from the date of entry into the study. Here events are defined as:

    - Non achievement of hematologic response including CR, CRi, CRh

    - Hematologic relapse or progressive disease

    - Initiation of any subsequent anti-leukemic therapy (excluding hydroxyurea). *Reasons for subsequent therapy initiation (including intolerance, resistance, hematologic relapse, unsatisfactory MRD evolution), its type (intensive, less-intensive, best supportive care) and its frame (clinical trial, off-label or compassionate use) will be prospectively recorded.*

    - Deaths prior to one of the events mentioned above.

  - Relapse free survival defined only for patients achieving CR, CRh or CRi and measured from the date of achievement of a remission

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- Duration of response
- Early mortality rates at day 28 (after 1 cycle of treatment) and day 56 (after 2 cycles)

- Incidence of subsequent allogeneic HSCT, overall and in responding patients specifically

7.2 Anticipated methods and timetable for measuring, collecting and analysing the efficacy data

The efficacy secondary endpoints will be assessed at C2D1, C3D1, and every three cycles and at EOI.

Criteria of response are assessed from peripheral blood count and bone marrow aspirates performed according to this timeline.

- Morphological examination, will be performed in local hematology laboratories.
- Eight color FCM MRD will be performed centrally in Saint Louis Hospital on bone marrow aspirates, only for patients achieving CR/CRi/CRh.
- In *NPM1* mutated patients, standard PCR-based MRD will also be conducted locally on the bone marrow and peripheral blood, only for patients achieving CR/CRi/CRh.
- In t(8;21) and inv(16) patients, standard PCR-based MRD will also be conducted locally on the bone marrow, only for patients achieving CR/CRi/CRh.

Samples for MRD, biomarkers and PD (25 ml of blood and 10 ml of bone marrow aspirates in EDTA tubes) should be sent at room temperature within 24 hours to the *Laboratoire d'Hématologie Hôpital Saint Louis 1 avenue Claude Vellefaux 75010 Paris*.

PK samples (5 mL of blood) will be collected into heparinized tubes. Immediately after collection, tubes will be inverted several times and kept at approximately 4 °C until centrifugation. The tubes will be centrifuged for 10 min at 1500×g at 4 °C within 60 min of collection to separate plasma. The plasma will be separated into 2 aliquots in polypropylene screw-cap tubes and placed at -80°C until analysis. Samples will be sent in dry ice to the Service de génomique des tumeurs et pharmacologie *Hôpital Saint Louis 1 avenue Claude Vellefaux 75010 Paris*.

8 SPECIFIC STUDY COMMITTEES

Two committees will govern the study.

English name	*French name*
Data Safety Monitoring Board (DSMB)	*Comité de surveillance indépendant (CSI)*
Steering Committee	*Comité de Pilotage*

8.1 Steering Committee

- Committee members:

The steering committee should meet the main players of the study:

- The coordinator investigator (Dr Lina Benajiba)

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- The biostatistician in charge of the project (Dr Lucie Biard, Saint-Louis, Unité de Recherche Clinique, Department of Biostatistics , Saint Louis hospital)
- The pharmacologist in charge of PK analysis (Dr Lauriane Goldwirt, Saint Louis pharmacology department, Paris)
- The principal investigators of each participating centers
- The scientific director and co-directors of the study (Pr Raphaël Itzykson, Dr Alexandre Puissant (INSERM U944, Institut de Recherche Saint-Louis, Paris), Dr Kris Wood (Duke University, USA)
- A representative of the sponsor (APHP DRCI)
- The coordination team of THEMA (Karine Celli-Lebras, Dr Renaud Buffet, Dr Laure Gilles).

- Roles:

The role of the steering committee is to:

- Define the overall structure of the study, coordinate information, determine the initial methodology and oversee the study.
- Propose procedures to be followed during the study, acknowledging any recommendations from the DSMB. The DRCI sponsor retains decision-making authority.
- At the end of the phase 1, determine in interaction with the DSMB the recommended phase 2 dose (RP2D).
- Review and validate the analysis plan of the data.
- Review and validate the draft of the publications.

8.2 DSMB

A 3-member Data Safety Monitoring Board (DSMB) will review safety data. *See* ***section 9.3.4.***

9 SAFETY ASSESSMENT - RISKS AND BURDEN ADDED BY THE STUDY

9.1 Description of Safety endpoints assessment parameters

Safety endpoints including DLT definition are described in **section 3.1.1** of the protocol.

Identification and grading of AEs is entirely based on NCI CTCAE version 5.0 of the NCI.

9.2 Anticipated methods and timetable for measuring, collecting and analysing the safety endpoints

Safety endpoints are easily monitored through clinical examination, usual laboratory tests and ECG.

During the phase I portion of the trial, **the eCRF should be filled in nearly real time** and monitored as soon as possible after the EOI visit. Real-time assessment of DLTs is central to the Bayesian continual reassessment method (CRM). All monitored DLTs will inform the modelling of the DL allocated to the next enrolled patient (see details on enrollment procedure in **Section 4.1.2** and on the CRM model in **Section 11**).

9.3 Recording and reporting adverse events

9.3.1 Definitions

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According to Article 2 of the Regulation (EU) No 536/2014:

- **Adverse event**

Any untoward medical occurrence in a subject, to whom a medicinal product is administered and which does not necessarily have a causal relationship with this treatment. .

- **Serious adverse event**

Any untoward medical occurrence that at any dose requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity, results in a congenital anomaly or birth defect, is life-threatening, or results in death.

- **Unexpected serious adverse reaction**

A serious adverse reaction, the nature, severity or outcome of which is not consistent with the reference safety information.

According to Article 53 of the Regulation (EU) No 536/2014:

- **Unexpected event**

An unexpected event which affect the benefit-risk balance of the clinical trial, but are not suspected unexpected serious adverse reactions as referred to in Article 42. That notification shall be made without undue delay but no later than 15 days from the date the sponsor became aware of this event.

According to Article 54 of the Regulation (EU) No 536/2014:

- **Urgent safety measure**

Where an unexpected event is likely to seriously affect the benefit-risk balance, the sponsor and the investigator shall take appropriate urgent safety measures to protect the subjects. The sponsor shall notify the Member States concerned, through the EU portal, of the event and the measures taken.
That notification shall be made without undue delay but no later than seven days from the date the measures have been taken.

9.3.2 The role of the investigator

The investigator must **assess the seriousness of each adverse event** and record all serious and non-serious adverse events in the case report form (eCRF).
The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the intensity** of the adverse events: by using the' Common Terminology Criteria for Adverse Events (CTCAE of the National Cancer Institute), version 5.0 of November 2017.

The investigator must **assess the causal relationship** between serious adverse events and the investigational medicinal products capivasertib and selinexor.

The method used by the investigator is based on 2 causality terms (EVCTM method):

- Related
- Not related

9.3.2.1 Serious adverse events that require the investigator to notify the sponsor without delay

The investigator notifies the sponsor without undue delay but not later than within 24 hours on the day the investigator becomes aware of any serious adverse event which occurs during a study that meets the description in article 41 of Regulation (EU) N°536/2014,

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with the exception of any event which is listed in the protocol and, if applicable, in the investigator's brochure as not requiring notification.

A serious adverse event is any untoward medical occurrence that:

1- results in death
2- is life-threatening to the participant enrolled in the study
3- requires hospitalisation or prolongation of existing hospitalisation
4- results in persistent or significant disability/incapacity
5- is a congenital anomaly/birth defect

Medication errors, pregnancies and uses outside what is foreseen in the protocol, including misuse and abuse of the product, require the investigator to notify the sponsor without delay.

9.3.2.2 Specific features of the protocol

9.3.2.2.1 Other events that require the investigator to notify without delay the sponsor

- Adverse events deemed "medically significant"
    - Any grade ≥ 3 non hematological adverse events fitting with DLT criteria.

The investigator must notify the sponsor without delay on the day the investigator becomes aware of these adverse events, in the same manner and within the same deadline as for serious adverse events (see above).

- ***In utero* exposure**

The investigator must notify the sponsor without delay on the day the investigator becomes aware of any pregnancy that occurs during the study, even if it is not associated with an adverse event.
If the investigational medicinal product is genotoxic, every case of maternal or paternal exposure must be notified.

- Exposure **while breastfeeding**

Exposure while breastfeeding occurs if an infant or a child may have been exposed to a medicinal product *via* the breast milk of a mother being treated with an investigational medicinal product.
Even if it is not associated with an adverse event, the investigator must notify the sponsor without delay on the day the investigator becomes aware of the exposure while breastfeeding.

9.3.2.2.2 Serious adverse events that do not require the investigator to notify the sponsor without delay

These serious adverse events are only recorded in the case report forms. A data retrieval of the case report forms will be implemented for serious adverse events at EOI at the latest.

- *Normal and natural course of the condition*:
    - Hematological toxicities except if persisting at grade ≥3 after day 42 in the absence of proven disease persistence or underlying myelodysplastic syndrome by bone marrow aspiration and/or biopsy.
- *Special circumstances:*
    - Hospitalisation for a pre-existing illness or condition
    - Hospitalisation for a medical or surgical treatment scheduled prior to the study
    - Admission for social or administrative reasons

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- *Adverse events potentially related to authorized auxiliary medicinal products and related to treatments prescribed as part of the care provided during the study follow-up*
  The investigator must report these adverse events to the relevant regional pharmacovigilance centre, Centre Régional de Pharmacovigilance (CRPV).

9.3.2.3 Period during which SAEs must be notified without delay by the investigator to the sponsor

The investigator must notify the sponsor without delay of any serious adverse events as defined in the corresponding section:

- from the date on which the participant begins treatment with capivasertib and selinexor,
- throughout the whole follow-up period required for the trial,
- until 4 weeks after the end of the participant's treatment with capivasertib and selinexor,
- indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g. serious reactions that could appear long after exposure to the medication, such as cancers or congenital abnormalities).

9.3.2.4 Procedures and deadlines for notifying the sponsor

The initial notification of an SAE must be provided in a written report signed by the investigator using a SAE notification form specific to the study and intended for this purpose (in the case report form).

Each item in this form must be completed by the investigator so that the sponsor can carry out the appropriate analysis.

The initial notification to the sponsor of a serious adverse event must be quickly followed by an additional detailed written report (or reports) so that the case outcome may be monitored by the Safety Department or to provide further information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful (medical reports, laboratory test results, results from additional examinations, etc.). These documents must be non-identifying. In addition, the documents must include the following: study acronym, number and participant’s initials.

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the participant has left the study.

The initial notification, the SAE follow-up reports and all other documents must be sent to the sponsor's Safety Department by email (eig-vigilance.drc@aphp.fr). It should be noted that it is possible to send SAE reports to the Safety Department by fax to +33 (0)1 44 84 17 99 only in the event of a failed attempt to send the SAE report by email (in order to avoid duplication).

For studies which use e-CRFs

- the investigator completes the SAE notification form in the e-CRF, then validates, prints, and signs the form before sending it by email;
- if it is not possible to connect to the e-CRF, the investigator should complete, sign, and send the SAE notification form to the Safety Department. As soon as the connection is restored, the SAE notification form in the e-CRF must be duly completed.

The investigator must respond to all requests from the sponsor for additional information.
For all questions relating to the notification of an adverse event, the Safety Department can be contacted via email: vigilance.drc@aphp.fr.

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For cases of *in utero* exposure, the investigator will complete the Notification and Follow-up form for a pregnancy occurring during participation in a study".
The investigator must monitor the pregnant woman throughout her pregnancy or until the pregnancy is terminated, and must notify the sponsor of the outcome of the pregnancy using this form.
If the outcome of the pregnancy falls within the definition of a serious adverse event (miscarriage, pregnancy termination, foetal death, congenital abnormality, etc.), the investigator must follow the procedure for reporting SAEs.
The initial pregnancy notification, the SAE follow-up reports, and any other documents will be sent to the sponsor according to the same procedures specified herein.
If it was the father who was exposed, the investigator must obtain the pregnant woman's permission before collecting information about the pregnancy.

9.3.3 Role of the sponsor

The sponsor, represented by its Safety Department, shall continuously, throughout the trial, assess the safety of each investigational medicinal product throughout the study.

9.3.3.1 Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all the adverse events reported
- the **causal relationship** between these events and each investigational medicinal product and any other treatments,
  All serious adverse events which the investigator and/or the sponsor reasonably believe may have a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.
- the **expected or unexpected nature** of the serious adverse reactions
  Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorised, is considered unexpected.
  The sponsor, represented by its Safety Department, assesses the expected/unexpected nature of a serious adverse reaction based on the information described below**.**

❖ For serious adverse events likely to be related to the investigational medicinal product selinexor:
- refer to the SmPC for selinexor that will be enclosed in CTIS platform.

❖ For serious adverse events likely to be related to the investigational medicinal product capivasertib:
- refer to the Investigator's Brochure that will be enclosed in CTIS platform.

❖ The serious adverse events potentially related to the interventions, procedures or examinations specific to the study are:
- For blood samples: hematoma, pain, vagal malaise, mild bleeding
- For ECG: mild cutaneous eruption, mild irritation
- For bone marrow aspiration: bleeding, pain, hematoma

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs) via Eudravigilance, within the regulatory time frame, to the competent authority:

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- in the case of fatal or life-threatening suspected unexpected serious adverse reactions, as soon as possible and in any event not later than seven days after the sponsor became aware of the reaction;
- in the case of non-fatal or non-life-threatening suspected unexpected serious adverse reactions, not later than 15 days after the sponsor became aware of the reaction;
- in the case of a suspected unexpected serious adverse reaction which was initially considered to be non-fatal or nonlife threatening but which turns out to be fatal or life-threatening, as soon as possible and in any event not later than seven days after the sponsor became aware of the reaction being fatal or life-threatening
- in the uncompleted case of fatal or life-threatening suspected unexpected serious adverse reactions, all additional information to complete this case have to be reported as soon as possible and in any event not later than eight days after the initial report
- All additional, relevant information must be declared by the sponsor in the form of follow-up reports within a period of 15 days starting from when the sponsor had this information.

The sponsor must notify all the investigators involved about any information that could adversely affect the safety of the research participants.

Specific case of serious adverse events of special interest:
The sponsor may be required to declare serious adverse events of special interest, in accordance with the same procedures and deadlines as for SUSARs.

9.3.3.2 Analysis and declaration of other safety data

This means any new information that prompts a reassessment of the risk/benefit ratio of the trial or of the product under investigation, or which could be sufficient to consider changes to the use of the product, the trial procedures or trial documentation, or the suspension, cancellation or amendment of the research trial or of other similar trials. For trials involving the first administration or use of a health product in healthy volunteers: any serious adverse reaction.

The sponsor will report in CTIS platform and to ANSM without delay upon knowledge of any emerging safety issue and, if applicable, describe what measures have been taken.

Following the initial declaration of any emerging safety issue, the sponsor will address to competent authorities any additional relevant information about the emerging safety issue in the form of a follow-up report, which must be sent no later than 8 days after learning of the information.

9.3.3.3 Annual safety report

Once a year for the duration of the clinical study, the sponsor must produce an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:
- a safety analysis for the research participants,
- a description of the patients included in the study (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- summary tables of all the serious adverse events that have occurred since the start of the study.

The sponsor produces one annual safety report (Development Safety Update Report - DSUR) for one clinical trial.

The report must be submitted in CTIS no later than 60 days after the anniversary of the date on which the competent authority authorised the trial.

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9.3.4 Data Safety Monitoring Board (DSMB)

The Data Safety Monitoring Board (DSMB) may be established by the sponsor. Its primary mission is to monitor safety data. It can have other missions, such as monitoring efficacy data (especially if the protocol includes interim analyses).
The sponsor is responsible for justifying the creation or absence of such a committee to the Competent Authority and to the Ethics Committee).

A DSMB will be established for this trial. The DSMB must hold its first meeting before the first participant is enrolled and ideally before the protocol is submitted to the competent authority and the Ethics Committee).
All missions as well as the specific operating procedures of the DSMB are described in the study's DSMB charter.

The DSMB members are (To be confirmed):
- Two experts in AML
- One methodologist

The DSMB will operate in accordance with the sponsor's procedures. The DSMB works in an advisory capacity only and the sponsor retains all decision-making authority.

10 DATA MANAGEMENT

10.1 Data collection procedures

Data collection in a centralized online platform for Case Report Forms (CRFs) will be performed by the investigating teams in all participating centers, under the responsibility of the principal investigators.
Case Report Forms (CRFs) and related source documents will be reviewed in detail during monitoring visit (completeness, adherence to the guidelines, accuracy compared to source documents). The sponsor's representative will also review regulatory documents, drug storage and accountability.
The investigator must maintain a comprehensive and centralized filing system of all study-related documentation that is suitable for inspection by sponsor's monitors or representatives of other regulatory agencies.
The investigator will allow sponsor's representatives to monitor the study at the frequency defined in the contract, depending on enrolment at each center.

10.2 Identification of data recorded directly in the CRFs which will be considered as source data

Not applicable

10.3 Right to access data and source documents

10.3.1 Data access

In accordance with GCPs and appendix 1 of the European Regulation N°536-2014:
- the sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source

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documents and the reports, for the purposes of the sponsor's quality control and audit procedures or inspections by the competent authority
- the Sponsor declares that investigators and participating institution will ensure the persons in charge of monitoring, quality control and auditing or inspecting the clinical trial have access to the documents and personal data strictly necessary for these tasks, in accordance with the statutory and regulatory provisions in force

10.3.2 Source documents

Source documents are defined as any original document or item that can prove the existence or accuracy of a data or a fact recorded during the study. These documents will be kept in accordance with the regulations in force by the investigator or by the hospital in the case of a hospital medical file.

In the context of the study source document are medical files, original biological examination results, summary from imaging examinations.

10.3.3 Data confidentiality

The persons responsible for the quality control of clinical studies will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the study, the study participants and in particular their identity and the results obtained.
These persons, as well as the investigators themselves, are bound by professional secrecy.
During and after the clinical trial, all data collected concerning the participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be rendered non-identifying.
Under no circumstances shall the names and addresses of the participants involved be shown. *Only the participant's initials will be recorded, accompanied by an encoded number specific to the study indicating the order of enrolment.*
The sponsor will ensure that each participant has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

10.4 Data processing and storage of research documents and data

10.4.1 Identification of the data processing manager and location(s)

Dr. Lucie BIARD from the Clinical Research Unit (URC, an INCa labeled Data Center) and Service Biostatistique et Informatique Médicale (SBIM) of Hôpital Saint Louis, AP-HP, Paris will be responsible for data entry and the relevant procedures. The same goes for conducting the statistical analysis.

10.4.2 Data entry

Non-identifying data will be entered electronically via a web browser.

10.5 Data ownership

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

11 STATISTICAL ASPECTS

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11.1 Description of statistical methods to be used including the timetable for the planned interim analyses

Study samples:

- Intention to treat sample: Includes all included subjects. This will refer to the primary analyses
- Per protocol sample: Includes all subjects from the intent-to-treat set without any major violations which could affect the evaluation of the primary efficacy endpoint. This will be used as sensitivity analyses.

Intent-to-treat set will be used for all main analyses. Per protocol set will be used for sensitivity analyses.

As a general strategy, continuous endpoints will be summarized using summary measures (median and interquartile range). Frequency distributions (counts and percentages) will be used to summarize categorical endpoints. Similarly, characteristics of patients will be presented using summary measures (median and interquartile range for quantitative characteristics and counts and percentages for categorical characteristics).
Time-to-event endpoint will be summarized using Kaplan Meier estimates.

Phase I - Primary analysis: The **primary estimand** will be the dose-limiting toxicity rate at 28 days, in patients with in relapsed/refractory acute myeloid leukemia receiving the combination Selinexor/capivasertib at any of the candidate doses combination. The primary endpoint will be the posterior mean probability of DLT at each of the candidate combination dose levels. A product of independent beta probabilities dose escalation phase I design will be used in phase 1, for up to 24 patients [Mander & Sweeting 2015]. Target DLT rate for the definition of the MTD contour (MTC) will be 20%, across the dosing grid including 8 candidate drug doses combinations, defined by 3 candidate dose levels of Selinexor and 4 of Capivasertib, including 4 non-tested combinations that will be set non-admissible *a priori* (see section Treatments). A time-to-event extension design may be used depending on the anticipated accrual rate compared to the DLT observation window (observation window 28 days) [Wheeler et al 2019]. Analysis of the DLT probability will be unadjusted, and sequential, upon initiation of any new dose-finding cohort.

Phase II – Primary analysis: A two-stage Bayesian optimal phase II design (BOP2) will be used, assessing the probability of complete response and monitoring the risk of limiting toxicities within the first 28 days after treatment initiation [Zhou et al 2017], with an interim analysis with futility/toxicity stopping rules. The **primary estimand** will be the complete response and the limiting toxicity at 28 days, in patients with in relapsed/refractory acute myeloid leukemia receiving the combination Selinexor/capivasertib at the RP2D.
Specifically, let n denote the interim sample size, and N denote the maximum sample size. Let Y_{eff} and Y_{tox} denote the binary efficacy and toxicity endpoints, with $Y_{eff} = 1$ and $Y_{tox} = 1$ indicating that patients experience efficacy and toxicity, respectively. We assume that the joint distribution of (Y_{eff}, Y_{tox}) follows a multinomial distribution with four elementary outcomes: (Y_{eff}, Y_{tox}) = (1, 1), (1, 0), (0, 1) and (0, 0). Let $\boldsymbol{p} = (P_{11}, P_{10}, P_{01}, P_{00})$ denote the probabilities of observing the four outcomes, and let $p_{eff} = Pr(Y_{eff} = 1)$, $p_{tox} = Pr(Y_{tox} = 1)$ and $p_{efftox} = Pr(Y_{eff} = 1, Y_{tox} = 1)$. When efficacy and toxicity endpoints are monitored separately, the joint distribution reduced to marginal distribution of efficacy and marginal distribution of toxicity, respectively. This optimization is performed assuming a vague Dirichlet prior $Dir(0.04,0.06,0.36,0.54)$ for $\boldsymbol{p}$.
The **co-primary endpoints** will be posterior mean probabilities of 28-day CR and of limiting toxicity, using this vague prior. They will be reported with their 95% credible interval.

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11.2 Calculation hypotheses for the number of participants required and the result

Phase I
A product of independent beta probabilities dose escalation phase I design will be used in phase 1, for up to 24 patients (27). Target DLT rate for the definition of the MTD contour (MTC) will be 20%, across the dosing grid including 8 candidate drug doses combinations, defined by 3 candidate dose levels of Selinexor and 4 of Capivasertib, including 4 non-tested combinations that will be set non-admissible *a priori* (see section Treatments). A time-to-event extension design may be used depending on the anticipated accrual rate compared to the DLT observation window (observation window 28 days) (28).
The design will be calibrated in a comprehensive simulation study (beta priors for each of the combinations, escalation strategy, cohort size and safety constraints) to ensure coherence of the dose escalation scheme and optimize operating characteristics (notably, probability of correct MTC selection, probability of overdose selection, safety of treatment assignments during the trial).

Phase II:
Once phase 1 has been completed providing the estimated MTD, an extension of up to 44 patients will be initiated. Assuming 24 patients have been included in the phase 1 part of the study, 20 new patients will be included in the phase 2 part.
The phase II part will follow a Bayesian adaptive design, assessing the probability of response (overall response rate) and monitoring toxicity (29). A maximum total of 20 patients, with an interim analysis on the first 9 observations with futility and toxicity decision rules, allows 80% power given a 5% one-sided false positive rate (31), assuming a 10% response rate as unacceptable and 40% as desirable, while a limiting toxicity rate of 40% will be considered as unacceptable and 20 % as acceptable (as in the phase 1 part). We assume independent response and toxicity rates. Decision rules are detailed in **section 11.4.**

11.3 Anticipated level of statistical significance

The primary endpoints of this phase I/II study will be analysed using Bayesian approaches.
The phase I main analysis relies on the probability of DLT estimate at each doses combinations and not on any statistical testing. The phase II part of the study will follow a Bayesian design (see above, **section 11.2**), which will be calibrated to control the risk of false positive under 5% (31).
Analyses of the secondary endpoints will be performed in a descriptive manner.

11.4 Statistical criteria for termination of the study.

For the phase II part, using the BOP2 design, a maximum total of 20 patients, with an interim analysis on the first 9 observations with futility and toxicity decision rules, allows 80% power given a 5% one-sided false positive rate (31), assuming a 10% response rate p_{eff} as unacceptable and 40% as desirable, while a limiting toxicity rate $p_{tox}$of 40% will be considered as unacceptable and 20 % as acceptable (as in the phase 1 part). We assume independent response and toxicity rates.

The treatment will be deemed unacceptable if $p_{eff} \leq 0.1$ or $p_{tox} > 0.4$. Thus, we will stop enrolling patients and claim that the treatment is unacceptable if

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$$Pr(p_{eff} > 0.1|data) < \lambda\left(\frac{n}{N}\right)^{\alpha},$$

or

$$Pr(p_{tox} \leq 0.4|data) < \lambda\left(\frac{n}{N}\right)^{\alpha/3},$$

where λ=0.72 and α=0.74 are design parameters optimized to maximize the study power, i.e., probability of correctly concluding an efficacious and safe treatment as acceptable when $p_{eff} = 0.4$, $p_{tox} = 0.2$ and $p_{efftox} = 0.08$, while controlling that the probability of incorrectly claiming an inefficacious and toxic treatment, i.e., type I error, with $p_{eff} = 0.1$, $p_{tox} = 0.4$ and $p_{efftox} = 0.04$, to 5%.
In other words, among the first 9 patients, if no response has been observed (no. response=0) or if 4 or more limiting toxicities have been observed, the trial will be stopped. Otherwise (if at least 1 response and no more than 3 limiting toxicities), accrual will resume up to a maximum of 20 patients in this part. At the second analysis, we will reject the undesirable hypothesis and conclude to a promising treatment if ≥4 responses are observed overall with ≤6 toxicities.

11.5 Method for taking into account missing, unused or invalid data

Missing data will be described.
Missing data on the primary endpoints will be imputed by a treatment failure for the primary analysis. Sensitivity analyses will be performed on complete cases for primary endpoints estimation.
For the secondary endpoints, analysis will be firstly performed on the intent to treat sample. Missing values on endpoints will be handled by simple imputation (for continuous endpoints: last observation carried forward when available or mean value otherwise, for binary endpoints: failure). Sensitivity analyses will be performed upon various hypotheses on missing values.

11.6 Management of modifications made to the analysis plan for the initial strategy.

All modifications of the initial plan will be tracked and recorded in the statistical analysis plan, and submitted to the scientific committee and the sponsor.

11.7 Choice of individuals to be included in the analyses

Intent-to-treat set will be used for all main analyses. Per protocol set will be used for sensitivity analyses.
Subgroup analyses according to the TP53 mutation status will be performed for the secondary endpoints.

12 QUALITY CONTROL AND ASSURANCE

Every clinical trial sponsored by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored clinical trial.

12.1 General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the study. The sponsor must implement a quality assurance system to best monitor the implementation of the study in the investigation centres.

For this purpose, the sponsor shall assign Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study locations, after having carried out the initial visits.

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The purpose of monitoring the study, as defined in Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

12.1.1 Strategy for centre opening

The strategy for opening the centres established for this study is determined using the appropriate monitoring plan.

12.1.2 Scope of centre monitoring

In the case of this level D risk study the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact, resulting in a study monitoring level to be implemented: level D.

12.2 Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical Research and Innovation Department) and in accordance with Good Clinical Practices as well as the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits carried out by the Clinical Research Associate. During these visits, the following elements will be reviewed depending on the monitoring level:

- written consent
- compliance with the study protocol and with the procedures defined therein
- quality of the data collected in the case report form: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used

12.3 Case report forms

All information required by the protocol must be entered in the case report forms. The data must be collected as and when they are obtained, and clearly recorded in these case report forms. Any missing data must be coded.
Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given a document offering guidance on using this tool.
When the investigators complete the case report form via the Internet, the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, the data are immediately verified as they are entered, thanks to consistency checks. To this end, the investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment.
The investigator must archive a copy of the authenticated document that was issued to the sponsor.

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12.4 Management of non-compliances

Any events that occur as a result of non-compliance – by the investigator or any other individual involved in running the study – with the protocol, standard operating procedures, good clinical practices or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.

These non-compliances will be managed in accordance with the sponsor's procedures.

The sponsor shall notify the Member States concerned about a serious breach of this Regulation or of the version of the protocol applicable at the time of the breach through the EU portal without undue delay but not later than seven days of becoming aware of that breach.

12.5 Audits/inspections

The investigators agree to consent to the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the grounds of medical secrecy.

An audit can be carried out at any time by individuals appointed by the sponsor and independent of those responsible for the research. The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.

The persons who manage and monitor the study agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results, including the storage of the data used or produced as part of the study.

12.6 Principal Investigator's commitment to assume responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal *curriculum vitæ*, signed and dated less than one year, with his/her RPPS number (*Répertoire Partagé des Professionnels de Santé*, Collective Database of Health Professionals, for France). The CV must include any previous involvement in clinical research and related training.
Any conditions, such as economic interests and institutional affiliations that might influence the impartiality of the investigators shall be presented.

Each investigator will commit to comply with legislation and to conduct the study in line with GCP, in accordance with the Declaration of Helsinki. Each investigator will give the sponsor's representative a GCP certificate dated fewer than three years ago.

The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCI document) which will be sent to the sponsor's representative.
The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

12.7 Suitability of the facilities

Before starting the study, each clinical trial sites will give the sponsor's representative a duly justified written statement on the suitability adapted to the nature and use of the investigational

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medicinal product and including a description of the suitability of facilities, equipment, human resources and description of expertise.

12.8 Pharmacist's commitment of responsibility

Production problems must be recorded according to the Manufacturing unit (LTCG) procedures.

13 ETHICAL AND LEGAL CONSIDERATIONS

13.1 Methods for informing research participants and obtaining their consent

According to article 29 of European regulation N°536/2014, No clinical trials on medicinal products for human use can be carried out on a person without his/her freely given and informed consent, obtained in writing after the person has been given the information specified in article 29.2 of the European Regulation.

A reflection period of at least 24 hours is given to the individual between the time when he or she is informed and when he or she signs the consent form.

The person's freely-given, written, informed consent will be obtained by the principal investigator or a physician representing the investigator before the person is enrolled in the study.

A copy of the information note and consent form, signed and dated by the research participant and by the principal investigator or the physician representing the investigator will be given to the individual prior to their participation in the study. The principal investigator or the physician representing him/her will keep a copy.

At the end of the study, one copy will be placed in a tamper-proof sealed envelope containing all the consent forms. This envelope will be archived by the sponsor.

In addition, the investigator will specify in the person's medical file the person's participation in the research, the procedures for obtaining his/her consent as well as the methods used for providing information for the purpose of collecting it. The investigator will retain one copy of the signed and dated consent form.

13.2 Prohibition from participating in another clinical study

The participant may not enrol in another interventional study for the duration of his or her participation without consulting with the physician monitoring him or her in the context of the study.

The participants can however participate in other non-interventional studies.

13.3 Compensation for participants

No compensation for participants is planned.

13.4 Registration on the National Register of study participants to clinical trial concerning the products mentionned in Article L. 5311-1 of the Code de la santé publique

13.5 Authorisation for the research location

In France, the study will be carried out in treatment units on individuals presenting a clinical condition for which the units are specialised and who require interventions that are routinely

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performed at those units. Therefore, the research location does not require specific authorisation.

13.6 Legal obligations

13.6.1 Role of the sponsor

Assistance Publique - Hôpitaux de Paris (AP-HP) is the sponsor of this study and, by delegation, the DRCI (Clinical Research and Innovation Department) carries out the study's missions in accordance with regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and all national Laws. Assistance Publique - Hôpitaux de Paris reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

13.6.2 Request for authorisation l

Prior to starting the study, AP-HP, as sponsor, must obtain authorisation from the Competent Authority and the Research Committee for this clinical trial on medicinal product for human use within the scope of its authority and in accordance with in force legislation and regulatory requirements.

13.6.3 Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended “Informatique et Libertés” law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.

The research is governed by the “Reference Methodology” (MR-001).

- Commitment to comply with “Reference Methodology” MR-001

This research is governed by the CNIL (French Data Protection Agency) “Reference Methodology for processing personal data used within the scope of health research” (amended MR-001). AP-HP, as sponsor of the research, has signed a declaration of compliance with this “Reference Methodology”.

13.6.4 Start of the Clinical Trial

The sponsor shall notify each Member State concerned of the start of a clinical trial and the start of the recruitment through the EU portal. That notification shall be made within 15 days from the start of the clinical trial.

13.6.5 Amendments to the Clinical Trial

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain approval from the Ethics Committee) and authorisation from the competent authority within the scope of their respective authorities, before the amendment can be implemented.
The information note and the consent form can be revised if necessary, in particular in case of a substantial amendment to the study or if adverse reactions occur.

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13.6.6 End of the Clinical Trial

The end of the Clinical Trial corresponds to the end of the participation of the last person who participate to the Clinical Investigation.
The end of the recruitment of subjects for the clinical trial and the end of the clinical trial should be notified in CTIS. That notification shall be made within 15 days from the end of the clinical trial in relation to that Member State.

13.6.7 Summary of the results of the clinical trial

According to article 37.4 of the European regulation n°536/2014, irrespective of the outcome of a clinical trial, within one year from the end of a clinical trial in all Member States concerned, the sponsor shall submit to the CTIS a summary of the results of the clinical trial. This summary shall be accompanied by a summary written in a manner that is understandable to laypersons.

13.6.8 Archiving

Specific documents for a clinical trial on medicinal product for human use will be archived by the investigator and the sponsor for 25 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- A sealed envelope for the sponsor, containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list):
  - the successive versions of the protocol (identified by the version number and its date), and any appendices
  - the competent authority authorisations and Research Ethics Committee decisions
  - any correspondence
  - the enrolment list or register
  - the appendices specific to the research
  - final study report
- The data collection documents

14 FUNDING AND INSURANCE

14.1 Funding sources

The study will be funded by the *PHRC-I (Hospital Funding for Clinical Research*). Drugs will be provided by Astrazeneca (Capivasertib) and Menarini (Selinexor). The study is developed in the THEMA program (ANR IHU-B-2018) and the resources of the THEMA program (study coordination) will contribute to the study conduction.

14.2 Insurance

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study

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to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with HDI-GLOBAL SE through the insurance broker BIOMEDIC-INSURE for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with Article 76(1) of regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and Article L.1121-10 of the *Code de la Santé Publique* (French Public Health Code).

15 PUBLICATION RULES

The author(s) of any publication relating to this study must include the AP-HP among their affiliations and must name the sponsor AP-HP (DRCI) and the source of funding. A copy of the publication must be sent to the sponsor (see below for rules governing affiliation, and naming of the sponsor and funders).

15.1 Mention of AP-HP affiliation for projects sponsored by AP-HP

- If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is unimportant
- Each of these affiliations must be identified by an address and separated by a semicolon (;)
*- The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: **AP-HP**, hospital, department, city, postcode, France*

15.2 Mention of the sponsor AP-HP (DRCI) in the acknowledgements of the text

- "The sponsor was Assistance Publique – Hôpitaux de Paris (Direction de la Recherche Clinique et de l'Innovation)"

15.3 Mention of the financial backer in the acknowledgements of the text

The study will be funded by a grant from Programme Hospitalier de Recherche Clinique Inter-régional – PHRC-I 20xx (French Ministry of Health).

This study will be registered on the website http://clinicaltrials.gov/ under number xxx.

16 BIBLIOGRAPHY

1. Visser O, Trama A, Maynadié M, Stiller C, Marcos-Gragera R, De Angelis R, et al. Incidence, survival and prevalence of myeloid malignancies in Europe. Eur J Cancer. 2012 Nov;48(17):3257–66.

2. Dombret H, Gardin C. An update of current treatments for adult acute myeloid leukemia. Blood. 2016 Jan 7;127(1):53–61.

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3. Othus M, Appelbaum FR, Petersdorf SH, Kopecky KJ, Slovak M, Nevill T, et al. Fate of patients with newly diagnosed acute myeloid leukemia who fail primary induction therapy. Biol Blood Marrow Transplant. 2015 Mar;21(3):559–64.

4. Koenig K, Mims A, Levis MJ, Horowitz MM. The Changing Landscape of Treatment in Acute Myeloid Leukemia. Am Soc Clin Oncol Educ Book. 2020 Mar;40:1–12.

5. Dombret H, Itzykson R. How and when to decide between epigenetic therapy and chemotherapy in patients with AML. Hematology Am Soc Hematol Educ Program. 2017 Dec 8;2017(1):45–53.

6. DiNardo CD, Pratz K, Pullarkat V, Jonas BA, Arellano M, Becker PS, et al. Venetoclax combined with decitabine or azacitidine in treatment-naive, elderly patients with acute myeloid leukemia. Blood. 2019 Jan 3;133(1):7–17.

7. Sébert M, Renneville A, Bally C, Peterlin P, Beyne-Rauzy O, Legros L, et al. A phase II study of guadecitabine in higher-risk myelodysplastic syndrome and low blast count acute myeloid leukemia after azacitidine failure. Haematologica. 2019 Aug;104(8):1565–71.

8. Chari A, Vogl DT, Gavriatopoulou M, Nooka AK, Yee AJ, Huff CA, et al. Oral Selinexor-Dexamethasone for Triple-Class Refractory Multiple Myeloma. N Engl J Med. 2019 Aug 22;381(8):727–38.

9. Etchin J, Sanda T, Mansour MR, Kentsis A, Montero J, Le BT, et al. KPT-330 inhibitor of CRM1 (XPO1)-mediated nuclear export has selective anti-leukaemic activity in preclinical models of T-cell acute lymphoblastic leukaemia and acute myeloid leukaemia. Br J Haematol. 2013 Apr;161(1):117–27.

10. Etchin J, Montero J, Berezovskaya A, Le BT, Kentsis A, Christie AL, et al. Activity of a selective inhibitor of nuclear export, selinexor (KPT-330), against AML-initiating cells engrafted into immunosuppressed NSG mice. Leukemia. 2016 Jan;30(1):190–9.

11. Ranganathan P, Yu X, Na C, Santhanam R, Shacham S, Kauffman M, et al. Preclinical activity of a novel CRM1 inhibitor in acute myeloid leukemia. Blood. 2012 Aug 30;120(9):1765–73.

12. Garzon R, Savona M, Baz R, Andreeff M, Gabrail N, Gutierrez M, et al. A phase 1 clinical trial of single-agent selinexor in acute myeloid leukemia. Blood. 2017 Jun 15;129(24):3165–74.

13. Bhatnagar B, Zhao Q, Mims AS, Vasu S, Behbehani GK, Larkin K, et al. Selinexor in combination with decitabine in patients with acute myeloid leukemia: results from a phase 1 study. Leuk Lymphoma. 2020 Feb;61(2):387–96.

14. Sweet K, Komrokji R, Padron E, Cubitt CL, Turner JG, Zhou J, et al. Phase I Clinical Trial of Selinexor in Combination with Daunorubicin and Cytarabine in Previously Untreated Poor-Risk Acute Myeloid Leukemia. Clin Cancer Res. 2020 Jan 1;26(1):54–60.

15. He Y, Sun MM, Zhang GG, Yang J, Chen KS, Xu WW, et al. Targeting PI3K/Akt signal transduction for cancer therapy. Signal Transduct Target Ther. 2021 Dec 16;6(1):425.

16. Banerji U, Dean EJ, Pérez-Fidalgo JA, Batist G, Bedard PL, You B, et al. A Phase I Open-Label Study to Identify a Dosing Regimen of the Pan-AKT Inhibitor AZD5363 for Evaluation

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in Solid Tumors and in PIK3CA-Mutated Breast and Gynecologic Cancers. Clin Cancer Res. 2018 May 1;24(9):2050–9.

17. Smyth LM, Tamura K, Oliveira M, Ciruelos EM, Mayer IA, Sablin MP, et al. Capivasertib, an AKT Kinase Inhibitor, as Monotherapy or in Combination with Fulvestrant in Patients with AKT1 E17K-Mutant, ER-Positive Metastatic Breast Cancer. Clin Cancer Res. 2020 Aug 1;26(15):3947–57.

18. Yap TA, Kristeleit R, Michalarea V, Pettitt SJ, Lim JSJ, Carreira S, et al. Phase I Trial of the PARP Inhibitor Olaparib and AKT Inhibitor Capivasertib in Patients with BRCA1/2- and Non-BRCA1/2-Mutant Cancers. Cancer Discov. 2020 Oct;10(10):1528–43.

19. Kolinsky MP, Rescigno P, Bianchini D, Zafeiriou Z, Mehra N, Mateo J, et al. A phase I dose-escalation study of enzalutamide in combination with the AKT inhibitor AZD5363 (capivasertib) in patients with metastatic castration-resistant prostate cancer. Ann Oncol. 2020 May;31(5):619–25.

20. Lin KH, Rutter JC, Xie A, Killarney ST, Vaganay C, Benaksas C, et al. P2RY2-AKT activation is a therapeutically actionable consequence of XPO1 inhibition in acute myeloid leukemia. Nat Cancer. 2022 Jul;3(7):837–51.

21. Emdal KB, Palacio-Escat N, Wigerup C, Eguchi A, Nilsson H, Bekker-Jensen DB, et al. Phosphoproteomics of primary AML patient samples reveals rationale for AKT combination therapy and p53 context to overcome selinexor resistance. Cell Rep. 2022 Aug 9;40(6):111177.

22. Sykes SM, Lane SW, Bullinger L, Kalaitzidis D, Yusuf R, Saez B, et al. AKT/FOXO signaling enforces reversible differentiation blockade in myeloid leukemias. Cell. 2011 Sep 2;146(5):697–708.

23. Lindsley RC, Mar BG, Mazzola E, Grauman PV, Shareef S, Allen SL, et al. Acute myeloid leukemia ontogeny is defined by distinct somatic mutations. Blood. 2015 Feb 26;125(9):1367–76.

24. Balsat M, Renneville A, Thomas X, de Botton S, Caillot D, Marceau A, et al. Postinduction Minimal Residual Disease Predicts Outcome and Benefit From Allogeneic Stem Cell Transplantation in Acute Myeloid Leukemia With NPM1 Mutation: A Study by the Acute Leukemia French Association Group. J Clin Oncol. 2017 Jan 10;35(2):185–93.

25. Jourdan E, Boissel N, Chevret S, Delabesse E, Renneville A, Cornillet P, et al. Prospective evaluation of gene mutations and minimal residual disease in patients with core binding factor acute myeloid leukemia. Blood. 2013 Mar 21;121(12):2213–23.

26. Döhner H, Wei AH, Appelbaum FR, Craddock C, DiNardo CD, Dombret H, et al. Diagnosis and management of AML in adults: 2022 recommendations from an international expert panel on behalf of the ELN. Blood. 2022 Sep 22;140(12):1345–77.

27. Mander AP, Sweeting MJ. A product of independent beta probabilities dose escalation design for dual-agent phase I trials. Stat Med. 2015 Apr 15;34(8):1261–76.

28. Wheeler GM, Sweeting MJ, Mander AP. A Bayesian model-free approach to combination therapy phase I trials using censored time-to-toxicity data. J R Stat Soc Ser C Appl Stat. 2019 Feb;68(2):309–29.

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Version 4.0 dated 31/05/2019

29. Zhou H, Lee JJ, Yuan Y. BOP2: Bayesian optimal design for phase II clinical trials with simple and complex endpoints. Stat Med. 2017 Sep 20;36(21):3302–14.

30. FDA (US Food & Drug Administration). Codevelopment of Two or More New Investigational Drugs for Use in Combination. 2013.

31. FDA (Food & Drug Administration). Adaptive designs for clinical trials of drugs and biologics. 2019.

32. Lo-Coco F, Avvisati G, Vignetti M, Thiede C, Orlando SM, Iacobelli S, et al. Retinoic acid and arsenic trioxide for acute promyelocytic leukemia. N Engl J Med. 2013 Jul 11;369(2):111–21.

33. Hoxhaj G, Manning BD. The PI3K-AKT network at the interface of oncogenic signalling and cancer metabolism. Nat Rev Cancer. 2020 Feb;20(2):74–88.

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17 LIST OF ADDENDA

17.1 List of investigators

Address of the study location	Title	First name Surname	Telephone/E-mail/Fax
Hôpital Saint-Louis, Paris	Dr	Lina Benajiba	Tel: +33 1 42 38 52 70 Mail: lina.benajiba@aphp.fr
Hôpital Saint-Louis, Paris	Pr	Raphaël Itzykson	Tel: +33 1 42 38 51 27 Mail: raphael.itzykson@aphp.fr
Gustave Roussy, Villejuif	Dr	Stéphane de Botton	Tel. +33 1 42 11 40 79 Email:stephane.debotton@gustaveroussy.fr
Institut Paoli Calmettes	Dr	Sylvain Garciaz	Tel: +33 4 91 22 36 95 Email: garciazs@ipc.unicancer.fr
CHU Nice, Nice	Pr	Thomas Cluzeau	Tel: +33 4 92 03 58 41 Email: cluzeau.t@chu-nice.fr
Hôpital Avicenne, Bobigny	Pr	Thorsten Braun	Tel: +33 1 48 95 70 51 Email: thorsten.braun@aphp.fr

17.2 Serious Adverse Events notification form

Direction de l'Organisation Médicale et des relations avec les Universités (DOMU) Direction de la Recherche Clinique et de l'Innovation (DRCI)	ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS **Formulaire de notification d'un Evènement Indésirable Grave (EIG) survenant au cours d'une recherche impliquant la personne humaine portant sur un Médicament ou produit assimilé**	**PARTIE RESERVEE AU PROMOTEUR** **Version 9 – 23/06/2022** **REFERENCE VIGILANCE :** Référence GED : REC-DTYP-0192

Dès la prise de connaissance de l'EIG par l'investigateur, ce formulaire doit être dûment complété (4 pages), signé et retourné sans délai au secteur Vigilance de la DRCI par mail (eig-vigilance.drc@aphp.fr)

Il est possible de transmettre les formulaires de notification d'EIG au secteur Vigilance par télécopie au +33 (0)1 44 84 17 99 uniquement en cas de tentative infructueuse d'envoi par mail afin d'éviter les doublons.

Notification initiale ☐ **Suivi d'EIG** ☐ **N° du suivi** |__|__|

1. Identification de la recherche

Acronyme :

Code de la Recherche :

Essai portant sur la 1ère administration du médicament chez des personnes qui ne présentent aucune affection (à supprimer si non applicable)

Date de notification : |__|__| |__|__| |2|0|__|__| (jj mm aaaa)

Date de prise de connaissance de l'EIG par l'investigateur : |__|__| |__|__| |2|0|__|__| (jj mm aaaa)

Titre complet de la Recherche :

Risque : ☐ A ☐ B ☐ C ☐ D

Plan expérimental : ☐ Essai non comparatif ☐ Essai comparatif ☐ Double aveugle ☐ Simple aveugle ☐ Ouvert ☐ Randomisé ☐ Non randomisé

2. Identification du centre investigateur

Nom de l'établissement : .....
Ville et code postal : .....
Service : .....

Investigateur (nom/prénom) : .....
Tél : ..... Fax : .....
Mail : .....

3. Identification et antécédents de la personne se prêtant à la recherche

Référence de la personne : |__|__|__| - |__|__|__|__|__| - |__| - |__| (n°centre - n° ordre de sélection - initiale nom - initiale prénom)

Sexe : ☐ M ☐ F

Date de naissance :

Antécédents médicaux-chirurgicaux/familiaux pertinents pour l'évaluation du cas (joindre un CRH anonymisé le cas échéant) :

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Poids : |__|__|__| kg Taille : |__|__|__| cm	|__|__| |__|__|__|__| mm aaaa Age : |__|__|__|ans	.....
Date de signature du consentement : |__|__| |__|__| |_2_|_0_|__|__| jj mm aaaa Date d'inclusion : |__|__| |__|__| |_2_|_0_|__|__| jj mm aaaa		
Date de randomisation : |__|__| |__|__| |_2_|_0_|__|__| jj mm aaaa		☐ Groupe (à compléter*) ☐ Groupe (à compléter*) *préciser de manière explicite si les personnes concernées ne présentent aucune affection N° randomisation (si applicable) : N° traitement (si applicable) :

4. Médicament(s) <u>expérimental(aux)</u> (ME) ou produit(s) assimilé(s) [préciser le(s)quel(s)] avant la survenue de l'EIG
(barrer l'encadré si traitement non débuté)

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie(1)	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours(2)	Date de fin *(jj/mm/aaaa)*
.....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|
.....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|

5. Médicament(s) <u>auxiliaire(s) nécessaire(s) à la réalisation de la recherche : si applicable</u> [préciser le(s)quel(s)] avant la survenue de l'EIG ***(barrer l'encadré si traitement non débuté)***

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie(1)	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours(2)	Date de fin *(jj/mm/aaaa)*
.....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|
.....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|

PARTIE RESERVEE AU PROMOTEUR
Version 9
REFERENCE VIGILANCE :
Référence GED : REC-DTYP-0185

Acronyme :
Référence de la personne se prêtant à la recherche : \|_\|_\|_\| - \|_\|_\|_\|_\|_\| - \|_\|_\| - \|_\|
n°centre **n° ordre de sélection** **initiale nom** **initiale prénom**

6. Procédures et actes ajoutés par la recherche ***(ex. : biopsies, IRM …)*** ***(barrer l'encadré si procédures et actes non réalisés)***	Date de réalisation *(jj/mm/aaaa)*	Chronologie	
		Avant la survenue de l'EIG	Après la survenue de l'EIG
.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	☐
.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	☐

7. Médicament(s) <u>concomitant(s)</u> au moment de l'EIG, à l'exclusion de ceux utilisés pour traiter l'événement indésirable ***(compléter le tableau ci-après et si nécessaire l'annexe relative aux médicaments concomitants ou barrer l'encadré si non applicable)***

⇨ **Annexe jointe au présent formulaire :** ☐ Oui ☐ Non

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie(1)	Posologie (préciser l'unité ex : mg/j)	Dates d'administration *(du jj/mm/aa au jj/mm/aa)*	En cours (2)	Indication	***Action prise*** *0 : poursuite sans modification de la posologie* *1 : arrêt* *2 : diminution de la posologie* *3 : augmentation de la posologie* *4 : ne sais pas*	***Causalité de l'EIG*** *0 : non lié au médicament* *1 : lié au médicament* *2 : ne sais pas*
			du |__|__| |__|__| |__|__| au |__|__| |__|__| |__|__|	☐			
			du |__|__| |__|__| |__|__| au |__|__| |__|__| |__|__|	☐			

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser) (2) En cours au moment de la survenue de l'EIG

8. Evènement indésirable grave [EIG]	
<u>Diagnostic</u> : ☐ Définitif ☐ Provisoire	**<u>Organe(s) concerné(s)</u> :**

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.....
.....
.....

Date de survenue des premiers symptômes : |__|__| |__|__| |_2_|_0_|__|__|
Préciser lesquels :
.....

.....
.....
.....
.....
.....
.....

Date d'apparition de l'EIG :
|__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa
Heure de survenue : |__||__| hh |__||__| min
☐ *donnée manquante*

Délai entre la date de la dernière administration du ME/produit assimilé ou la date de procédure/acte ajouté par la recherche et la date de survenue de l'EIG :
|__||__| / |__||__| |__||__|
jj hh min

Critères de gravité :

☐ Nécessite ou prolonge l'hospitalisation :
du |__|__| |__|__| |_2_|_0_|__|__|
au |__|__| |__|__| |_2_|_0_|__|__| ☐ *en cours*

☐ Décès
☐ Mise en jeu du pronostic vital
☐ Incapacité ou handicap important ou durable
☐ Anomalie ou malformation congénitale
☐ Autre(s) critère(s) médicalement significatif(s), préciser :
.....
.....

L'évènement a-t-il conduit à :
☐ aucune mesure prise concernant le ME
☐ diminution de la posologie du ME, préciser la nouvelle posologie : .....
date de début de la nouvelle posologie : |__|__| |__|__| |_2_|_0_|__|__|
☐ augmentation de la posologie du ME, préciser la nouvelle posologie : .....
date de début de la nouvelle posologie : |__|__| |__|__| |_2_|_0_|__|__|
☐ arrêt définitif du ME, préciser la date : |__|__| |__|__| |_2_|_0_|__|__|
☐ arrêt transitoire du ME, préciser la date : |__|__| |__|__| |_2_|_0_|__|__|
date de reprise : |__|__| |__|__| |_2_|_0_|__|__|
☐ ne sais pas

Récidive de l'EIG après ré-administration : □ Non □ Oui Date : |__|__| |__|__| |_2_|_0_|__|__|
□ Non applicable

Des mesures symptomatiques ont-elles été prises ?

☐ Non ☐ Oui Date : |__|__| |__|__| |_2_|_0_|__|__|Préciser : .....

.....

Une levée d'insu a-t-elle été nécessaire après la survenue de l'évènement ?
☐ Non ☐ Oui Date : |__|__| |__|__| |_2_|_0_|__|__| ☐ Non applicable

Degré de sévérité ***(à adapter à la recherche) :***
☐ Léger ☐ Modéré ☐ Sévère

Ou selon la classification CTCAE ***(à adapter à la recherche) :***

☐ Grade 1 ☐ Grade 2 ☐ Grade 3
☐ Grade 4 ☐ Grade 5

L'évènement fait-il suite à :

- une erreur médicamenteuse ?	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|
- un surdosage ?	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|
- un mésusage ?	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|
- autre (préciser) :	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|

PARTIE RESERVEE AU PROMOTEUR
Version 9
REFERENCE VIGILANCE :
Référence GED : REC-DTYP-0185

Acronyme :
Référence de la personne se prêtant à la recherche : |__|__|__| - |__|__|__|__| - |__| - |__|
n°centre **n° ordre de sélection** **initiale nom** **initiale prénom**

Evolution de l'événement

☐ **Décès**
- ■ sans relation avec l'EIG
- ■ en relation avec l'EIG

Date : |__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa

☐ **Sujet non encore rétabli**, préciser :
■ Etat stable ■ Amélioration ■ Aggravation

☐ **Résolu :**
- ■ sans séquelles
- ■ avec séquelles, préciser lesquelles :

Date : |__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa
|__|__| |__|__|
hh min

☐ **Evolution inconnue**

9. Autre(s) étiologie(s) envisagée(s)

☐ Non ☐ Oui Si oui, préciser : .....
.....
.....

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10. Examen(s) complémentaire(s) réalisé(s)
☐ Non ☐Oui Si oui, préciser date, nature et résultats : [joindre les bilans anonymisés]…… …… ……

11. Selon l'investigateur, l'événement indésirable grave est (plusieurs cases possibles)
Lié à la recherche : ☐ **Oui :** ☐ **au(x) médicament(s)/produit(s) assimilé(s) de la recherche : le(s)quel(s) ?** **Lequel : ……** ☐ **Lié** ☐ **Non lié** **Lequel : ……** ☐ **Lié** ☐ **Non lié** ☐ **au(x) médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche : le(s)quel(s) ?** **Lequel : ……** ☐ **Lié** ☐ **Non lié** **Lequel : ……** ☐ **Lié** ☐ **Non lié** ☐ **à la (aux) procédure(s)/acte(s) de la recherche : la/le(s)quel(les) ?** **La/lequel(le) : ……** ☐ **Lié** ☐ **Non lié** **La/lequel(le) : ……** ☐ **Lié** ☐ **Non lié** ☐ **Non :** ☐ **à la progression de la maladie faisant l'objet de la recherche : (à compléter)** ☐ **à un (ou plusieurs) médicament(s) concomitant(s) administré(s), le(s)quel(s) : ……** ☐ **à une maladie intercurrente, laquelle : ……** ☐ **autre, préciser : …….**

Notificateur	Investigateur	Tampon du service :
Nom et fonction : Signature	Nom : Signature	

17.3 Pregnancy notification form

Direction de l'Organisation Médicale et des relations avec les Universités (DOMU) Direction de la Recherche Clinique et de l'Innovation (DRCI)	ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS ***Formulaire de notification et suivi d'une grossesse apparue au cours d'une recherche impliquant la personne humaine portant sur un Médicament ou produit assimilé***	**PARTIE RESERVEE AU PROMOTEUR** **Version 7 – 23/06/2022** **REFERENCE INTERNE :** Référence GED : REC-DTYP-0185

Ce formulaire doit être dûment complété (3 pages), signé et retourné sans délai au secteur Vigilance de la DRCI par mail (eig-vigilance.drc@aphp.fr)

Il est possible de transmettre les formulaires de notification de grossesse au secteur Vigilance par télécopie au +33 (0)1 44 84 17 99 uniquement en cas de tentative infructueuse d'envoi par mail afin d'éviter les doublons.

1. Identification de la recherche	Notification initiale ☐ Suivi de notification ☐ N° du suivi |__|__|
Acronyme : Code de la recherche : Risque :	**Date de notification :** |__|__| |__|__| |2|0|__|__| (jj mm aaaa) **Date de prise de connaissance de la grossesse par l'investigateur :** |__|__| |__|__| |2|0|__|__| (jj mm aaaa)

Titre complet de la Recherche :

2. Identification du centre investigateur	
Nom de l'établissement : …… Ville et code postal : …… Service : ……	Investigateur (nom/prénom) : …… Tél : Fax : E-mail :

3. Identification de la personne présentant une grossesse

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Référence de la personne : |__|__|__| - |__|__|__|__|__| - |__| - |__|
n°centre - n° ordre de sélection - initiale nom - initiale prénom

Date de naissance : |__|__| |___|___|__|__| (mm-yyyy)
Date d'inclusion : |__|__| |__|__| |_2_|_0_|__|__|
Date de randomisation : |__|__| |__|__| |_2_|_0_|__|__|
Poids : |__|__|__| kg
Taille : |__|__|__| cm
Groupe de randomisation : ☐ (à compléter) ☐ (à compléter)
N° randomisation : …… N° traitement : ……
Date des dernières règles : |__|__| |__|__| |_2_|_0_|__|__|
Et/ou date début de grossesse : |__|__| |__|__| |_2_|_0_|__|__|

Cas particulier d'une exposition paternelle : ☐ Oui ☐ Non

Référence de la personne : |__|__|__| - |__|__|__|__|__| - |__| - |__|
n°centre - n° ordre de sélection - initiale nom - initiale prénom

Date de naissance : |__|__| |___|___|__|__| (mm-yyyy)
Date d'inclusion : |__|__| |__|__| |_2_|_0_|__|__|
Date de randomisation : |__|__| |__|__| |_2_|_0_|__|__|
Groupe de randomisation : ☐ (à compléter) ☐ (à compléter)
N° randomisation : …… N° traitement : ……

Expositions au cours de la grossesse :

Tabac :	☐ non	☐ oui (préciser nombre de paquets/année) :	☐ arrêt (préciser date) :	☐ poursuite
Alcool :	☐ non	☐ oui (préciser unités OH) :	☐ arrêt (préciser date) :	☐ poursuite
Drogue :	☐ non	☐ oui (préciser substance) :	☐ arrêt (préciser date) :	☐ poursuite

Autre (préciser) :

4. Antécédents maternels

Médicaux :

Chirurgicaux :

Obstétricaux : |__|__| geste |__|__| pare

Préciser si fausse couche, grossesse extra-utérine, interruption de grossesse (médicale ou volontaire), mort *in utero*, malformation congénitale, pathologie congénitale/néonatale non malformative, … (*nombre, date et nature/raison si applicable).*

5. Médicament(s) expérimental (aux) administré(s) ou non pendant la grossesse ou s'il s'agit une exposition paternelle

Nom commercial (de préférence) ou Dénomination Commune Internationale	Date de première administration Ou non administré	Date de dernière administration Ou en cours	Voie d'administration(1)	Posologie / 24h
	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		
	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser)

Acronyme :

PARTIE RESERVEE AU PROMOTEUR
Version 7
RÉFÉRENCE INTERNE :
REC-DTYP-0185

Référence de la personne : |__|__|__| - |__|__|__|__|__| - |__| - |__|
n°centre - n° ordre de sélection - initiale nom - initiale prénom

6 . Médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche administré(s) ou non pendant la grossesse ou s'il s'agit une exposition paternelle : si applicable

Nom commercial (de préférence) ou Dénomination Commune Internationale	Date de première administration Ou non administré	Date de dernière administration Ou en cours	Voie d'administration(1)	Posologie / 24h
	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		
	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		

7. Procédures et actes ajoutés par la recherche ***(Barrez l'encadré si procédures et actes non réalisés)***	Date de réalisation *(jj/mm/aaaa)*	Chronologie	
		Avant la grossesse	Au cours de la grossesse
	|__|__| |__|__| |_2_|_0_|__|__|		
	|__|__| |__|__| |_2_|_0_|__|__|		

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8. Médicament(s) concomitants administré(s) dans le cadre du soin

(Cf. annexe « Liste relative aux médicaments concomitants » complétée : ☐ ***Oui*** ☐ ***Non applicable)***

Nom commercial (de préférence) ou Dénomination Commune Internationale	Date de première administration	Date de dernière administration Ou en cours	Voie d'administration(1)	Posologie / 24h
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser)

9. Suivi de la grossesse

☐ Echographiques. Date(s) et résultats à préciser *(joindre les CR anonymisés)* :

☐ Autres examens. Date(s) et résultats à préciser *(joindre les CR anonymisés)* :

10. Grossesse en cours ☐ (envoyer par e-mail un nouveau formulaire complété à l'issue de la grossesse pour le suivi de la notification initiale)

ou issue de la grossesse ☐ (compléter ci-dessous)

Date : |__|__| |__|__| |_2_|_0_|__|__| Terme : |__|__| SA |__|__| J

☐ Fausse couche
→ Examen anatomo-pathologique disponible : ☐ Non ☐ Oui, précisez le résultat :

☐ Grossesse extra-utérine
→ Examen anatomo-pathologique disponible : ☐ Non ☐ Oui, précisez le résultat :

☐ Interruption de grossesse → Raison :
→ Examen anatomo-pathologique disponible : ☐ Non ☐ Oui, précisez le résultat :

☐ Accouchement : ☐ Spontané ☐ Provoqué ☐ Voie basse ☐ Césarienne

Naissance multiple : ☐ Non ☐ Oui, précisez le nombre :

Souffrance fœtale : ☐ Non ☐ Oui, précisez :

Mort-né : ☐ Non ☐ Oui, précisez :

Placenta normal : ☐ Oui ☐ Non, précisez :

Liquide amniotique : ☐ Clair ☐ Autre, précisez :

Anesthésie : ☐ Générale ☐ Péridurale ☐ Rachianesthésie ☐ Aucune

Acronyme :

PARTIE RESERVEE AU PROMOTEUR
Version 7
RÉFÉRENCE INTERNE :
REC-DTYP-0185

Référence de la personne : |__|__|__| - |__|__|__|__|__| - |__| - |__|
n°centre - n° ordre de sélection - initiale nom - initiale prénom

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11. Nouveau-né (Si naissance multiple, compléter les parties 1, 2, 3, 9 et 10 d'un nouveau formulaire et l'envoyer par e-mail)		
Sexe : ☐ Masculin ☐ Féminin Poids : |__|__|__|__| grammes Taille : |__|__|__| cm Périmètre crânien : |__|__|__| cm APGAR : 1 minute : _____ 5 minutes : _____ 10 minutes : _____		
Malformation(s) congénitale(s) : ☐ Non ☐ Oui, précisez :		
Pathologie(s) congénitale(s)/néonatale(s) non malformative(s) : ☐ Non ☐ Oui, précisez :		
Le nouveau-né a-t-il bénéficié d'un suivi particulier à la naissance : ☐ Non ☐ Oui, précisez : ☐ Non applicable		
Notificateur	**Investigateur**	Tampon du service :
Nom et fonction : Signature :	Nom : Signature :	

17.4 Therapy-related malignancy notification form

Direction de l'Organisation Médicale et des relations avec les Universités (DOMU) *Direction de la Recherche Clinique et de l'Innovation (DRCI)*	ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS ***Formulaire de notification des cancers secondaires/myélodysplasies survenant au cours d'une recherche impliquant la personne humaine portant sur un Médicament ou produit assimilé***	**PARTIE RESERVEE AU PROMOTEUR** ***Version 5 – 23/06/2022*** **REFERENCE VIGILANCE :** Référence GED : REC-DTYP-0191

Ce formulaire doit être dûment complété (4 pages), signé et retourné sans délai au secteur Vigilance de la DRCI par mail (eig-vigilance.drc@aphp.fr)

Il est possible de transmettre les formulaires de notification de cancers secondaires/myélodysplasies au secteur Vigilance par télécopie au +33 (0)1 44 84 17 99 uniquement en cas de tentative infructueuse d'envoi par mail afin d'éviter les doublons.

Notification initiale ☐ **Suivi de notification** ☐ **N° du suivi** |__|__|

1. Identification de la recherche	
Acronyme :	**Date de notification :** |__|__| |__|__| |2|0|__|__| j j m m a a a a
Code de la Recherche :	**Date de prise de connaissance du des cancer secondaire/myélodysplasie par l'investigateur :** |__|__| |__|__| |2|0|__|__| j j m m a a a a
Titre complet de la Recherche :	**Risque :** ☐ A ☐ B ☐ C ☐ D **Plan expérimental :** ☐ Essai non comparatif ☐ Essai comparatif ☐ Double aveugle ☐ Simple aveugle ☐ Ouvert ☐ Randomisé ☐ Non randomisé

2. Identification du centre investigateur	
Nom de l'établissement : ..... Ville et code postal : ..... Service :	Investigateur (nom/prénom) : ..... Tél : ..... Fax : ..... Mail : .....

3. Identification et antécédents de la personne se prêtant à la recherche	
Référence de la personne : |__|__|__| - |__|__|__|__| - |__| - |__| n°centre - n° ordre de sélection - initiale nom - initiale prénom	**Antécédents médicaux-chirurgicaux/familiaux pertinents pour l'évaluation du cas (joindre un CRH anonymisé le cas échéant) :** .....
Sexe : ☐ M ☐ F — Date de naissance :	.....

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Poids : |__|__|__| kg Taille : |__|__|__| cm	|__|__| |__|__|__|__| m m a a a a Age : |__|__|__|ans	.....
Date de signature du consentement : |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a Date d'inclusion : |__|__| |__|__| |_2_|_0_|__|__| jj mm aaaa		
Date de randomisation : |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a		☐ Groupe (à compléter) ☐ Groupe (à compléter) N° randomisation (si applicable) : N° traitement (si applicable) :

4. Diagnostic du cancer secondaire/de la myélodysplasie

4.1 Diagnostic clinique :

Date du diagnostic : |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a	Diagnostic final retenu : .....
Confirmation histologique : ☐ **Non** ☐ **Oui**	
Confirmation cytologique : ☐ **Non** ☐ **Oui**	
4.2 Grade : (précisez l'échelle de classification ex : TNM)	☐ Grade 0 ☐ Grade I ☐ Grade II ☐ Grade III ☐ Grade IV .....
4.3 Grade histologique	☐ Grade 0 ☐ Grade I ☐ Grade II ☐ Grade III ☐ Grade IV
4.4 Si autre classification, précisez :	
4.5 Antécédents médicaux pertinents : ☐ Non ☐ Oui, précisez :	

PARTIE RESERVEE AU PROMOTEUR
Version 5
REFERENCE VIGILANCE :

Référence GED : REC-DTYP-0185

Acronyme :

Référence de la personne se prêtant à la recherche : \|__\|__\|__\| - \|__\|__\|__\|__\| - \|__\| - \|__\|
n°centre - n° ordre de sélection - initiale - initiale

5. Médicaments/procédures de la recherche

5.1 Médicament(s) <u>expérimental(aux)</u> (ME) ou produit(s) assimilé(s) [préciser le(s)quel(s)] avant la survenue de l'EIG
(barrer l'encadré si traitement non débuté)

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie[1]	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours[2]	Date de fin *(jj/mm/aaaa)*
.....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|
.....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|

5.2 Médicament(s) <u>auxiliaire(s) nécessaire(s) à la réalisation de la recherche : si applicable</u> [préciser le(s)quel(s)] avant la survenue de l'EIG
(barrer l'encadré si traitement non débuté)

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie[1]	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours[2]	Date de fin *(jj/mm/aaaa)*
.....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|

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....	.....	.....	|__|__| |__|__| |_2_|_0_|__|__|	☐	|__|__| |__|__| |_2_|_0_|__|__|

5.3 Procédures et actes ajoutés par la recherche ***(ex. : biopsies, IRM …)*** ***(barrer l'encadré si procédures et actes non réalisés)***	Date de réalisation *(jj/mm/aaaa)*	Chronologie	
		Avant la survenue de l'EIG	Après la survenue de l'EIG
....	|__|__| |__|__| |_2_|_0_|__|__|	☐	☐
....	|__|__| |__|__| |_2_|_0_|__|__|	☐	☐

6. Précision de l'imputabilité de l'investigateur

6.1 Selon l'investigateur, l'événement indésirable grave (cancer secondaire/myélodysplasie) est (plusieurs cases possibles)

Lié à la recherche :

☐ **Oui** : ▪ **au(x) médicament(s)/produit(s) assimilé(s) de la recherche : Le(s)quel(s) ?**

Lequel : ..... ☐ Lié ☐ Non lié

Lequel : ..... ☐ Lié ☐ Non lié

▪ **au(x) médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche : Le(s)quel(s) ?**

Lequel : ..... ☐ Lié ☐ Non lié

Lequel : ..... ☐ Lié ☐ Non lié

▪ **à la (aux) procédure(s)/acte(s) de la recherche : La/le(s)quel(les) ?**

La/lequel(le) : ..... ☐ Lié ☐ Non lié

La/lequel(le) : ..... ☐ Lié ☐ Non lié

☐ **Non** : ▪ **à la progression de la maladie faisant l'objet de la recherche :** (à compléter)

▪ **à un (ou plusieurs) médicament(s) concomitant(s) administré(s), le(s)quel(s) :** .....

▪ **à une maladie intercurrente, laquelle :** .....

▪ **autre, préciser :** .....

6.2 La survenue de cet EIG est-elle susceptible d'être liée à un manque d'efficacité du ME ?	☐ **Non** ☐ **Oui**

7. Détails de la chimiothérapie administrée pour traiter la pathologie initiale (phase)

☐ Induction |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a	☐ Consolidation |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a
☐ Post greffe : renseignez la partie 6.2	☐ Maintenance |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a
☐ Autre :	☐ Interphase

PARTIE RESERVEE AU PROMOTEUR
Version 5
REFERENCE VIGILANCE :

Référence GED : REC-DTYP-0185

Acronyme :

Référence de la personne se prêtant à la recherche : \|__\|__\|__\| - \|__\|__\|__\|__\|__\| - \|__\| - \|__\|
n°centre - n° ordre de sélection - initiale - initiale

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7.1 Médicament(s) ou produit(s) assimilé(s) de chimiothérapie anticancéreuse ou de thérapie ciblée avant la survenue du cancer secondaire/de la myélodysplasie *(barrez l'encadré si aucun traitement débuté) :*

Nom commercial ou Dénomination Commune International	Date de première administration Ou non administré	Date de dernière administration Ou en cours (2)	Voie d'administration(1)	Posologie / 24h	Lien de causalité avec l'EIG *(Relation selon méthode OMS)*
	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ Non administré	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ En cours			☐ non lié ☐ Relation certaine ☐ Relation probable ☐ Relation possible ☐ Relation improbable
	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ Non administré	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ En cours			☐ non lié ☐ Relation certaine ☐ Relation probable ☐ Relation possible ☐ Relation improbable

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser)
(2) En cours au moment de la survenue de l'EIG

7.2 Greffe de cellules souches hématopoïétiques (CSH) pour le traitement de la pathologie initiale :

☐ Non ☐ Oui, précisez ci-dessous :

Date de la greffe : le |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a ☐ autogreffe ☐ allogreffe	Si allogreffe : Donneur : ☐ apparenté ☐ fichier volontaires / banque

Origine CSH : ☐ CSP ☐ Moelle osseuse ☐ Sang de cordon

Date de sortie d'aplasie : \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\| j j m m a a a a

7.3 Traitements de conditionnement de la greffe (immunosuppresseurs, irradiation corporelle, etc.) :

☐ Non applicable ☐ Applicable, précisez ci-dessous le schéma thérapeutique :

Nom commercial ou Dénomination Commune Internationale	Date de première administration	Date de dernière administration	Voie d'administration(1)	Posologie / 24h
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__|		
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__|		

8. Statut de la pathologie initiale à la date de survenue du cancer secondaire/de la myélodysplasie

(Joindre si possible les résultats du dernier myélogramme le cas échéant) **:**

☐ Rémission complète le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
☐ Rémission avec séquelles le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|, précisez les séquelles :
☐ Rémission partielle le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|, précisez :
☐ Stable depuis le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
☐ Maladie en progression
☐ Rechute depuis le\|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|

9. Traitement du cancer secondaire/de la myélodysplasie

9.1 Hospitalisation(s) :

Hospitalisation (1) du \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\| au \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
Hospitalisation (2) du \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\| au \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
Hospitalisation (3) du \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\| au \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|

9.2 Intervention chirurgicale : ☐ Non ☐ Oui, précisez ci-dessous :

Type d'intervention chirurgicale :	Date de l'intervention chirurgicale : |__|__| |__|__| |_2_|_0_|__|__|

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Acronyme :

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n°centre - n° ordre de sélection - initiale - initiale

9.3 Chimiothérapie : ☐ Non ☐ Oui, précisez ci-dessous :
Précisez le schéma thérapeutique, date(s) de début, les posologies et dates de fin si applicable :
9.4 Radiothérapie : ☐ Non ☐ Oui, précisez ci-dessous :

Précisez le schéma thérapeutique et les doses :	Date de début : |__|__| |__|__| |_2_|_0_|__|__|
	Date de fin : |__|__| |__|__| |_2_|_0_|__|__|

9.5 Traitement(s) adjuvant(s) : ☐ Non ☐ Oui, précisez ci-dessous :
9.6 Une greffe de CSH a été réalisée pour le traitement du cancer secondaire/de la myélodysplasie : ☐ Non ☐ Oui, précisez ci-dessous :

Date de la greffe : le |__|__| |__|__| |_2_|_0_|__|__| ☐ autogreffe ☐ allogreffe	Si allogreffe : Donneur : ☐apparenté ☐ fichier volontaires / banque

Origine CSH :	☐ CSP	☐ Moelle osseuse	☐ Sang de cordon

Date de sortie d'aplasie : |__|__| |__|__| |_2_|_0_|__|__|

10. Evolution du cancer secondaire/de la myélodysplasie

10.1 Etat actuel (hors décès)

☐ Rémission complète le |__|__| |__|__| |_2_|_0_|__|__|
☐ Rémission avec séquelles le |__|__| |__|__| |_2_|_0_|__|__|, précisez les séquelles :
☐ Rémission partielle le |__|__| |__|__| |_2_|_0_|__|__|, précisez :
☐ Stable depuis le |__|__| |__|__| |_2_|_0_|__|__|
☐ Maladie en progression
☐ Rechute depuis le|__|__| |__|__| |_2_|_0_|__|__|

10.2 Evolution fatale

Date du décès : |__|__| |__|__| |_2_|_0_|__|__|

Autopsie effectuée : ☐ Non ☐ Oui (joindre le compte-rendu)

Veuillez spécifier la « cause du décès » rapportée dans le certificat de décès / le rapport d'autopsie :

Notificateur	**Investigateur**	Tampon du service :
Nom et fonction :	Nom :	

17.5 SmPC or Investigator's Brochure

SmPC for selinexor available on: http://agence-prd.ansm.sante.fr/php/ecodex/index.php

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17.6 List of CYP3A4 inhibitors and inducers.

Strong CYP3A4 inhibitors	**Moderate CYP3A4 inhibitors**	**Strong CYP3A4 inducers**	**Moderate CYP3A4 inducers**
boceprevir	amprenavir	apalutamide	bosentan
clarithromycin	aprepitant	carbamazepine	efavirenz
conivaptan	atazanavir	enzalutamide	etravirine
grapefruit juice	ciprofloxacin	mitotane	phenobarbital
indinavir	darunavir	phenytoin	primidone
itraconazole	diltiazem	rifampin	
ketoconazole	erythromycin	St. John's wort	
lopinavir	fluconazole		
mibefradil	fosamprenavir		
nefazodone	imatinib		
nelfinavir	verapamil		
posaconazole			
ritonavir			
saquinavir			
telaprevir			
telithromycin			
voriconazole			
Star fruit			
Seville oranges /products			

Note: The above list is not exhaustive. Also refer to:
https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers

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